

IMPERIAL

MENG INDIVIDUAL PROJECT

IMPERIAL COLLEGE LONDON

DEPARTMENT OF COMPUTING

**Beyond Before-and-After: Process-Quality
Evaluation of Refactoring Sessions TODO:**
add stuff about the paid tool + user
qualitative interviews in appendix

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May 29, 2026

Abstract

This report presents a tool for evaluating refactoring as a process, not just as a change between two endpoints. It defines a process-quality score for refactoring trajectories and uses that score to find points where a developer’s path diverges from a higher-scoring alternative. Each divergence is classified by kind, and the system builds a concrete counterfactual reference trajectory for comparison. Developer sessions are captured with an IntelliJ plugin, and the resulting scores and divergences are shown in a dashboard.

The system is evaluated on 45 hand-recorded refactoring sessions on a Java codebase, a 30-session randomised-arm user study on a separate codebase (five participants split into a with-feedback arm of three and a no-feedback baseline arm of two), and an 8-stack, 48-session agent extension on the same fixture as a feasibility experiment for the tool’s human-edit-pattern detectors. On the hand-recorded sessions, the tool achieves per-kind precision of 1.00 against inter-rater-agreed labels, with Cohen’s $\kappa \geq 0.72$ across the three rater pairs for all four divergence kinds. The score formula’s top-1 ranking remains stable under single-knob weight perturbations in 96.5% of cases on the user-study rankable subset (25 of 30 sessions), and in 84.6% of cases under simultaneous multi-knob Monte Carlo perturbation. The user-study arm comparison shows a randomised but underpowered direction-of-effect: across the six-session arc the with-feedback arm gains +21.0 on the gain-stripped process score and drives total divergence points from 12 to 2, while the no-feedback baseline arm loses 5.0 and rises from 10 to 15 divergence points. The agent extension finds no equivalent feedback effect, consistent with structural biases of the human-trained detectors against agent edit patterns.

Contents

1	Introduction	5
1.1	Background	5
1.2	Motivation	5
1.3	Research Objectives	6
1.4	Contributions	7
1.5	Approach and Scope	7
1.6	Thesis Roadmap	8
2	Background	9
2.1	Definitions and framing	9
2.1.1	Refactoring as behaviour-preserving transformation	9
2.1.2	Atomic refactorings, smells, and sequences	9
2.1.3	Representing refactoring as states and actions	10
2.2	Outcome-based refactoring evaluation	10
2.2.1	Judging refactoring by the final state	10
2.2.2	What endpoint evaluation captures-and what it misses	10
2.3	Code smell detection and quality metrics	11
2.3.1	Using code smells as design signals	11
2.3.2	Structural and readability metrics	11
2.3.3	What this means for the process-quality metric	11
2.4	Refactoring detection and mining from code changes	13
2.4.1	From diffs to refactoring actions	13
2.4.2	What refactoring detectors miss	13
2.4.3	Tie-in to our work	14
2.5	Refactoring recommendation and search/synthesis	14
2.5.1	Recommendation and synthesis as a contrast	14
2.5.2	Recent trajectory-formalism work in software engineering	15
2.6	Process-aware work and IDE trace capture	15
2.6.1	Logging developer interactions in IDEs	15
2.6.2	Granularity, segmentation, and semantic interpretation	16
2.7	Summary and positioning: toward process-quality evaluation	16
3	Methodology	18
3.1	Formal definitions and notation	18
3.2	Computing the process score	19
3.2.1	The process score	19
3.2.2	Process-score weights	20
3.2.3	The cleanliness sub-score	22
3.2.4	Event capture	24
3.2.5	Shadow repo reconstruction and worktree pool	24
3.2.6	Metrics calculation	24

3.3	Divergence points	25
3.3.1	Applying refactorings via Eclipse JDT	26
3.3.2	Ordering	26
3.3.3	Manual-Refactor	28
3.3.4	Rework	29
3.3.5	Hygiene	30
3.3.6	Validator and behaviour preservation	31
3.4	System architecture and pipeline phases	31
3.4.1	Architecture at a glance	31
3.4.2	Phase A / Phase B split	32
3.5	Dashboard	32
3.6	Score-formula evaluation design	33
3.6.1	Sensitivity perturbations	33
3.6.2	Ablation design	34
3.7	Evaluation dataset and labelling protocol	34
3.7.1	The 45-session injection set	35
3.7.2	Two label streams	35
3.7.3	Rater protocol	35
3.7.4	Cohen’s κ computation	35
3.7.5	Treatment of disagreements	36
3.7.6	Divergence-detector evaluation protocol	36
4	Experimentation and Evaluation	37
4.1	Testing the score formula	37
4.1.1	Sensitivity sweep	37
4.1.2	Ablation sweep	40
4.2	Testing the divergence-point detector	43
4.2.1	Inter-rater reliability	43
4.2.2	Divergence experiment	43
4.2.3	Known limitation: ordering recall	46
4.3	User and agent studies	46
4.3.1	User study	46
4.3.2	Agent extension experiment	51
5	Threats to validity	56
5.1	Construct validity	56
5.2	Internal validity	57
5.3	External validity	59
5.4	Statistical conclusion validity	61
5.5	Per-experiment threats	61
5.6	Implementation threats	62
5.7	Out of scope	63
6	Discussion and Conclusions	64
6.1	Summary of contributions	64
6.2	Headline findings	64
6.3	Discussion	65
6.4	Future work	66
6.5	Closing	66

7	Declarations	68
7.1	Use of Generative AI	68
7.2	Ethical Considerations	68
7.3	Sustainability	69
7.4	Availability of Data and Materials	69
A	Appendix	70

Chapter 1

Introduction

1.1 Background

Refactoring is a routine activity in software development in which developers restructure code to improve readability, maintainability, and evolvability while preserving externally observable behaviour; traditional definitions describe it as a sequence of small, semantics-preserving transformations [1, 2, 3], and subsequent surveys categorise these refactoring operations and the tools that support them [4]. In practice, refactoring is widely encouraged because poorly structured code becomes an obstacle when enhancing and maintaining code, and long-lived systems need continual adjustment (as argued in work building on Lehman’s laws and empirical studies of software maintenance). At the same time, refactoring can be costly: it consumes developer time, introduces risk, and is frequently intertwined with other work such as feature development and bug fixing rather than occurring as isolated “refactoring-only” tasks [5, 6].

1.2 Motivation

Most research and tooling that evaluates refactoring focuses on *outcomes* rather than the *process*. A common approach is to measure refactoring success as a change in static quality indicators between a “before” and “after” state: reductions in code smells, improvements in cohesion/coupling, decreases in complexity, or related metrics to do with maintainability. Typical strands of work include metric-driven detection or prioritisation of candidate refactorings (e.g., using metric changes to separate refactoring-like edits from functional ones, or to choose between different refactoring options), as well as refactoring recommendation systems that rank alternatives by their impact on design metrics. More recent work expands the signals used for guidance by incorporating traceability alongside code metrics when ranking refactoring solutions, and exploring how product/process metrics relate to developers’ motivations to refactor (e.g., studies building on mined refactorings and repository history, and proposals that use LLMs to determine developers’ motivations from commit artefacts). Alongside this, there is a substantial line of work that has improved the ability to *detect* refactorings from code changes, notably by using AST-based differencing and structured matching to infer refactoring operations between revisions (e.g., RefactoringMiner and related approaches), which has enabled large-scale empirical studies of how often refactoring happens, what kinds occur, and in what contexts [7, 8].

However, these perspectives evaluate refactoring as a mapping from an initial program state to a final program state, leaving the *trajectory* between those states largely outside the picture. This matters because developers rarely refactor in a single atomic operation: they work through intermediate states, sometimes take detours, redo work, and often accept temporary degradation (e.g., in readability, modular structure, or test/build stability)

before arriving at a final outcome. Two developers can arrive at the same end state via very different paths: one might take lots of disruptive steps, bounce between designs, or create avoidable churn, while another might get there via a steadier, more stable sequence. Existing endpoint-only evaluation cannot distinguish these different cases. Meanwhile, research on process realism suggests that refactoring is often “flossed” into ongoing development instead of being a separate dedicated task, which increases the chance of mixed-intent steps and noisy traces [5]. This makes the quality of the process (how a developer navigates intermediate states) potentially important for both developer productivity and the health of the codebase as it evolves.

Recent work has started to look more closely at refactoring trajectories themselves. RefactorBench, for example, uses a state/action/observation view to study how language-model agents fail on multi-file refactoring tasks [9], while Bouzenia et al. characterise recurring action patterns in software-engineering agent traces [10]. A recent systematic literature review also surveys how LLM-based refactoring work is evaluated and highlights several open problems in this area [11]. However, this still leaves a gap for evaluating an observed developer trace directly. The closest related systems either synthesise a path to a desired end state, without judging the path the developer actually took, or recommend refactorings towards an architectural target, without reference to an observed process. This project addresses that gap by scoring recorded refactoring traces with an explicit process-quality metric, generating higher-scoring counterfactual reference trajectories, and identifying measurable points where the observed trace diverges from those references.

1.3 Research Objectives

This thesis asks whether, given the start and end state of a refactoring session, we can judge the quality of the path the developer took and point to places where a better path was available. The core idea is to model refactoring as a *process* over states, rather than judging it only by comparing endpoints. To do this, we adopt a state/action/trace framing: snapshots of the codebase are treated as *states*, and developer edits—including IDE refactorings and manual transformations—are treated as *actions* that produce a trace from the start state to the end state.

A core component will be a **process-quality metric** that scores traces according to criteria that will be defined and justified using existing research on code quality metrics, smell detection validity, and process-level signals. The metric is not assumed to be purely structural: it may incorporate multiple terms capturing endpoint improvement, intermediate-state degradation, churn/disruption, and “safety” proxies such as preserving build and test behaviour, reflecting the known limitations of relying on any one signal alone.

To go beyond scoring a single observed refactoring trace we also construct *counterfactual* alternatives: **reference trajectories** between the same start and end states that score better under the chosen process-quality metric. These references give the observed trace something concrete to be compared against. The analysis can then identify *divergence points*, where the developer’s path moves away from a higher-scoring alternative, and estimate how much each divergence contributes to the final quality gap. This is related to systems that generate or recommend refactoring sequences: ReSynth synthesises a path to a target end state from a developer’s partial edits [12], Refactoring Navigator recommends steps towards an architectural target [13], and search-based refactoring treats sequences as an optimisation problem under explicit quality functions [14, 15]. The difference is that these systems are mainly concerned with producing a useful sequence. Here, the generated trajectory is used as a reference for evaluating the process the developer actually followed.

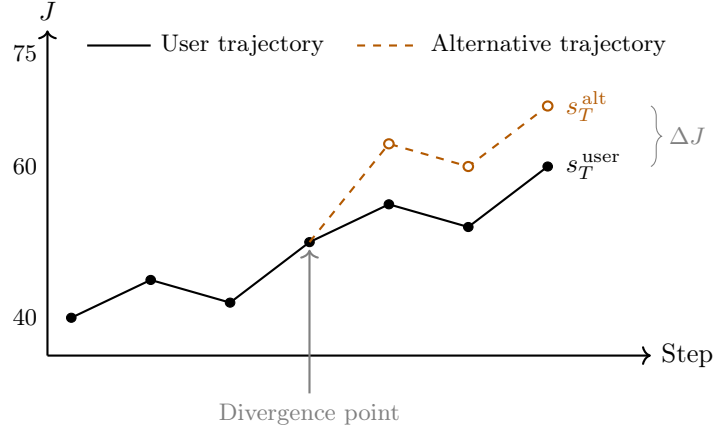


Figure 1.1: Illustration of two refactoring trajectories scored by the process-quality metric J . The solid line shows the developer’s observed trajectory, while the dashed line shows a higher-scoring reference trajectory that diverges at an intermediate state. The gap ΔJ is the estimated contribution of that divergence to the final score difference.

1.4 Contributions

The main contributions of the project are:

- A process-quality metric for refactoring trajectories. The metric combines terms from prior work on code quality and developer process signals – including both endpoint cleanliness gain and an intermediate-cleanliness lag penalty so that two trajectories which finish at the same code can still score differently if one front-loaded the cleanup – and we test how much the results change under sensitivity and ablation experiments.
- A divergence-point detector that compares an observed trace with a higher-scoring reference trajectory. The detector groups divergences into four kinds: *Ordering*, *Manual-Refactor*, *Rework*, and *Hygiene*.
- A 45-session dataset of recorded refactoring sessions on a purpose-built Java code-base. We provide multi-label ground-truth annotations and the experimental scripts used for the sensitivity, ablation, and divergence analyses.
- An empirical evaluation of the detector, reporting precision and recall for each divergence kind. The evaluation also shows which parts of the score formula are useful in practice, and which kinds of process improvement it does not capture well.

1.5 Approach and Scope

The approach is intended for realistic refactoring sessions, where IDE-supported refactorings and manual edits are often mixed together. In this project we focus on Java, since it gives us mature parsing tools, established quality metrics, and refactoring mining support such as RefactoringMiner. We represent a refactoring session as a sequence of code snapshots. Each step may correspond to a burst of manual edits or the use of an automated IDE refactoring tool, and the same model could later be applied to traces at different levels of granularity where that data is available.

The work does not attempt to enumerate every possible refactoring sequence or prove that a generated trajectory is globally optimal. Instead, it uses the analysis component to find a plausible higher-scoring reference path between the same start and end states.

This keeps the task focused on explaining the developer’s observed process, rather than replacing the developer with an automated refactoring planner.

The work is scoped to human developer sessions. The agent comparison in Chapter 4 is a feasibility extension on the same tooling rather than a co-equal evaluation target: the detectors are designed against human edit patterns (IDE refactor invocations, edit-burst structure, commit-cadence windows) and do not map cleanly onto agent traces. We treat that mismatch itself as one of the things Chapter 4 surfaces.

1.6 Thesis Roadmap

The remainder of the thesis is organised as follows. Chapter 2 reviews related work on refactoring evaluation, refactoring mining, and refactoring recommendation. Chapter 3 defines the process-quality metric and divergence-point detector. Chapter 4 describes the implementation of the tool. Chapter 5 presents the evaluation and discusses threats to validity. Chapter 6 concludes the thesis and outlines possible future work.

Chapter 2

Background

This chapter reviews the background needed to treat *refactoring as a process*, rather than only as a change between two endpoints. It begins with definitions of refactoring and refactoring sequences (§2.1), then discusses outcome-based evaluation and the code-quality signals commonly used to judge whether code has improved (§2.2–§2.3). It then reviews refactoring mining, recommendation and sequence synthesis, and IDE-based process tracing (§2.4–§2.6). The chapter ends by positioning this project against that work (§2.7).

2.1 Definitions and framing

2.1.1 Refactoring as behaviour-preserving transformation

Refactoring is usually defined as changing a program’s internal structure to improve qualities such as readability or maintainability, while preserving externally observable behaviour. Classic definitions stress that refactoring is built from small, semantics-preserving steps, with safety supported by preconditions and validation such as testing [1, 2, 3]. Opdyke’s work frames refactorings as program transformations with applicability conditions intended to preserve behaviour under a static approximation [1]. Fowler’s catalogue helped popularise refactoring as an incremental developer activity made up of many small operations [2, 3]. Mens and Tourwé later survey the wider research landscape, bringing together definitions, taxonomies, and tool support, and emphasising the role of precondition checking and program analysis in automated refactoring [4].

In practice, this clean definition is harder to apply. Developers often refactor while also adding features or fixing bugs, and some refactorings are only partly completed. As a result, it is often ambiguous whether a change seen in version history is a “pure refactoring”. In this thesis, behaviour preservation is therefore treated as an *intended property* of refactoring steps, while recognising that real traces may include edits with more than one purpose.

2.1.2 Atomic refactorings, smells, and sequences

Research on refactoring often describes refactorings as a set of *operators*, or atomic refactorings, such as EXTRACT METHOD, MOVE METHOD, and RENAME [2, 4]. Larger changes can then be understood as sequences of these smaller operations. Code smells are closely related: they are heuristic indicators of possible design problems, and therefore of possible refactoring opportunities [2, 3]. Although smells are not formal correctness criteria, they provide a useful bridge between developer judgement and measurable signals, discussed further in §2.3.

A key distinction in this thesis is between judging only the start and end states of a refactoring, and judging the path taken between them. Two developers may reach the same final code, but one path may involve more temporary degradation, churn, or disruption

than the other. These differences can matter for productivity and risk, even when the endpoint looks the same.

2.1.3 Representing refactoring as states and actions

To represent this refactoring path explicitly, we use a state/action/trace model. A codebase snapshot is treated as a *state* $s \in \mathcal{S}$, such as the repository at a particular point in time. An *action* $a \in \mathcal{A}$ is a developer edit step, which may be an IDE-supported refactoring, a manual change, or a mixture of both. A refactoring process is then a *trace*

$$\tau = (s_0, a_0, s_1, a_1, \dots, a_{T-1}, s_T),$$

where each action a_t transforms state s_t into s_{t+1} . This makes it possible to talk about “process quality” as a property of the whole trace τ , rather than only as a comparison between the first and final states (s_0, s_T) .

This framing is also close to recent software-engineering work that models agent and developer behaviour as a partially-observable Markov decision process, where a trajectory is a sequence of action and observation pairs [9]. We keep the explicit state notation because this thesis scores the intermediate code states themselves, not only the actions or observations produced along the way.

2.2 Outcome-based refactoring evaluation

2.2.1 Judging refactoring by the final state

Refactoring is often evaluated by comparing quality indicators before and after the change. Under this view, a refactoring is successful if the final snapshot improves on the starting snapshot according to selected measures, such as lower coupling, higher cohesion, lower complexity, or fewer code smells. This endpoint-focused view appears both in empirical studies of refactoring impact and in recommendation systems that rank candidate refactorings by their predicted improvement.

A clear example is *search-based refactoring*, where refactoring is treated as an optimisation problem over possible refactoring operations or sequences. Harman and Tratt apply multi-objective search at the design level, treating coupling and cohesion as objectives that may need to be traded off against each other [14]. Later work extends this idea to many-objective settings and adds constraints or preferences for practical concerns such as test coverage or prioritisation [15]. This line of work is useful here because it shows that “better” code is not a single fixed property: different quality goals can pull in different directions.

The same endpoint-based idea is also used in refactoring recommendation systems. These systems compare possible refactorings and rank them by the improvement they are expected to produce. Recent work has looked at how to choose among large sets of candidate refactorings, and which code characteristics are associated with positive or negative effects in practice [16].

2.2.2 What endpoint evaluation captures-and what it misses

Endpoint-based evaluation is useful when the main concern is whether the final code is better than the starting code under some metric Q . It gives recommendation systems a simple strategy: simulate each candidate refactoring and choose the one that gives the largest improvement in Q . It is also useful for empirical studies, where refactoring events can be compared with changes in measured code quality. This fits naturally with many existing metrics, since they are defined over a single codebase snapshot.

The weakness of this approach is that it says little about the refactoring process itself. Two developers may reach the same endpoint through very different paths, but endpoint metrics would treat those paths as equivalent. In practice, the intermediate states can still matter: code may become harder to understand for a period of time, the modular structure may be disrupted, or the build and tests may break before being repaired. The path can also be costly in other ways, for example by creating extra churn, making review harder, or increasing integration risk. Empirical work on modularity improvement shows this trade-off clearly: improvements in design metrics can come with substantial structural disruption [17]. This motivates evaluating not only the final code, but also the stability and cost of the path taken to reach it.

This is the point at which the process view becomes useful. Final-code metrics still matter, but they need to be considered alongside what happened on the way there: whether the work introduced temporary degradation, required rework, created unnecessary churn, or broke build/test stability.

2.3 Code smell detection and quality metrics

2.3.1 Using code smells as design signals

Code smells provide a way to connect qualitative design concerns with measurable indicators [2]. Smell detectors usually identify these cases either through hand-written rules or learned models. They often rely on structural properties such as size, coupling, and cohesion, with some tools also using lexical or semantic features. One prominent rule-based example is DECOR, which defines smells and design anti-patterns through measurable symptoms and detection rules [18]. DECOR is widely used in empirical studies because its definitions are explicit and it can be run efficiently at scale.

Smell detectors are useful, but they should not be treated as ground truth. Comparative studies show that tools can disagree in both precision and recall, and that performance varies across systems and smell types [19]. Much of this comes from the practical choices each tool makes, such as thresholds, feature sets, and implementation details. Detectors can also produce false positives, marking code as “smelly” even when there is no clear negative impact or when the design reflects an acceptable trade-off [20]. For this reason, smell counts are used in this thesis as one signal among several. They are combined with structural metrics, readability-oriented measures, and process-level safety signals, rather than being allowed to determine the process-quality score on their own.

2.3.2 Structural and readability metrics

Static metrics provide another way to measure code quality at the level of classes, methods, and packages. Common metric families cover coupling, cohesion, complexity, and size, and are often used as indicators of maintainability. These structural measures are useful, but they do not capture everything developers care about when refactoring. Readability is one example: models that combine structural features, such as line structure, with textual features, such as identifier patterns, tend to align better with human judgements of readability [21]. This matters for process evaluation because readability and understandability are common reasons for refactoring, and they can change noticeably in intermediate states even if the final code improves.

2.3.3 What this means for the process-quality metric

The process-quality metric $J(\tau)$ uses these signals, but it also has to account for their limitations. The literature suggests three design choices:

1. **Use multiple signals.** No single metric or smell detector gives reliable ground truth across all contexts [19, 20]. Combining several signals reduces dependence on the quirks of one tool.
2. **Separate endpoint improvement from process cost.** Improving the final snapshot is important, but disruption and temporary degradation can still matter even when the endpoint improves [17].
3. **Keep the components interpretable.** A process critique is more useful when it can point to a specific issue, such as a readability dip, churn spike, or smell increase, rather than only reporting a single opaque score.

A process metric should also include some notion of developer **effort**, such as the number of steps in the trajectory or the size of the edits between successive states. Fewer steps are not always better, since small steps can reduce risk, but step count and edit size are still useful for identifying unnecessary detours and rework. Similar cost terms are already used alongside structural quality goals in search-based refactoring and recommendation work, for example when trying to improve quality while minimising change [14, 15].

Table 2.1 summarises the candidate signals, their main benefits and limitations, and examples of Java tooling. The table does not define the final formula for $J(\tau)$, including weights or normalisation. Its purpose is to motivate the signals used later; the precise definition of $J(\tau)$ is given in the methodology chapter and used consistently in the evaluation.

Signal	Pros	Cons / threats	Typical tooling
Structural metrics (coupling/cohesion/complexity/size)	Widely used; efficient; interpretable at class/method granularity	Validity depends on context; may incentivise disruptive restructuring; metric interactions	Static analysis metric extractors; IDE/CI integrations
Smell counts / severity (rule-based)	Directly targets design issues; aligns with developer’s motivations	Detector disagreement and false positives; threshold sensitivity; context dependence	DECOR-style detectors [18]; other smell tools [19]
Readability models	Closer to human comprehension than structural-only metrics; captures lexical/style clues	Model generalisation and dataset bias; may be language/style dependent	Readability predictors [21]
Churn / disruption (LOC changed, file moves/renames)	Captures cost and instability of change; naturally trajectory-oriented	Churn may be necessary for large refactors; can penalise required restructuring	VCS mining; diff statistics
Process length / effort (number of steps; edit magnitude)	Trajectory-native cost signal; discourages detours/rework; easy to compute	Sensitive to step granularity (commits vs. checkpoints vs. IDE events); may penalise safe micro-steps; requires calibration	VCS checkpoints; IDE logs; diff/AST edit size
Build/test preservation signals	Direct safety signal; aligns with “behaviour-preserving intent”	Requires runnable builds/tests; noisy due to flaky tests; environment dependence	CI logs; local build/test runs

Table 2.1: Candidate signals for constructing a process-quality objective. We combine endpoint quality change with trajectory-oriented cost and stability terms, rather than relying on any single indicator.

2.4 Refactoring detection and mining from code changes

2.4.1 From diffs to refactoring actions

To evaluate a process trace in practice, developer actions either need to be captured directly, for example through IDE event logs, or inferred from the differences between snapshots such as bursts of manual user edits. Much of this work uses the second route: given two versions of a program, infer whether a refactoring happened and identify which refactoring types occurred.

Structured code differencing is important here because many refactorings are not easy to recognise from line-based diffs alone. AST-based differencing can represent moves, renames, and other non-local edits more directly. GumTree, for example, maps fine-grained AST nodes between program versions and produces edit scripts that better reflect the structural changes developers intended [8]. This kind of differencing is often used as the base layer for refactoring detection.

Modern refactoring miners use these structural mappings together with heuristics for particular refactoring types. RefactoringMiner detects a wide range of refactorings between two versions of the code by matching related classes, methods, and fields [7]. RefDiff takes a similar approach, using similarity heuristics and static analysis to detect common refactoring types in version histories [22]. For process analysis, tools like these are useful because they can turn a transition $s_t \rightarrow s_{t+1}$ into a set of detected refactoring actions, giving the trace a concrete *action vocabulary*.

2.4.2 What refactoring detectors miss

Refactoring detection tools usually report what kind of refactoring occurred, such as MOVE METHOD or EXTRACT CLASS, and which classes, methods, or fields were involved. This fits the operator-based view of refactoring used in refactoring guides and surveys [2, 4]. However, the literature also reports several limitations that matter for process evaluation.

Mixed and tangled changes. Version control commits often mix refactoring with feature work or bug fixes, and they may bundle several unrelated activities together. A single commit-level transition can therefore include both behaviour-preserving and behaviour-changing edits. In these cases, refactoring detection may return only partial or ambiguous results. This makes snapshot-derived actions harder to interpret and motivates careful trace construction, such as using finer checkpoints and methods that can tolerate imperfect refactoring detections.

Limited coverage and combined refactorings. Detectors only cover a finite set of operators, and refactorings can be composed, such as a move followed by a rename, in ways that make matching harder. Some refactorings are also performed manually without tool support and may not fit the patterns that detectors expect. For process evaluation, this means that “actions” should be treated as partially observed: detected refactorings are useful features, but they are not a complete description of developer intent.

Behaviour preservation is not verified. Detection is usually structural. It infers that a change *looks like* a refactoring, but it does not prove that behaviour was preserved. Because of this, downstream evaluation should not treat a detected refactoring as automatically safe. Where possible, it should also use independent safety signals, such as whether the build and tests still pass.

2.4.3 Tie-in to our work

Refactoring mining is important infrastructure for building traces and describing actions at scale [7, 22]. This thesis does not try to improve refactoring detection accuracy. Instead, mined refactorings are used as part of the observational layer: they provide labels and features for transitions between states, which can then support process scoring and reference-trajectory generation. The next section builds on this by discussing work that generates and navigates refactoring sequences.

2.5 Refactoring recommendation and search/synthesis

These systems generate paths; we evaluate observed ones. The literature on refactoring recommendation, sequence synthesis, and goal-directed navigation is therefore mainly relevant as a contrast: it establishes that refactoring can be treated as a structured sequence under explicit objectives, but the generated sequence is the object of interest rather than a developer’s actual trace. This section gives the contrast in one subsection; the more recent trajectory-formalism work that does engage with observed traces, which is closer to our framing, gets its own subsection.

2.5.1 Recommendation and synthesis as a contrast

Several strands of prior work show that refactoring sequences can be generated and optimised under explicit objectives. ReSynth, for example, synthesises a sequence of IDE-supported refactorings consistent with a developer’s partial edits, using the touched classes, methods, and fields together with operator preconditions to constrain the search [12]. Its goal is reachability rather than process quality, but it shows that a developer trace can provide useful evidence for constructing a refactoring path. Search-based refactoring takes a different route: Harman and Tratt frame refactoring as a multi-objective optimisation problem over design qualities such as cohesion and coupling, producing Pareto fronts rather than a single best answer [14]. Later many-objective work adds concerns closer to everyday development, including the number of refactorings performed [15] and architectural distance as an explicit disruption objective [23]. Refactoring Navigator is closer to a developer-facing recommender: it starts from a current system and a desired architecture, then recommends a sequence of refactoring steps using a search-based strategy [13]. Related empirical work on modularity optimisation also shows that improving structural metrics can substantially disturb the original modular structure [17], which motivates the disruption-aware penalties used later in §3.

This thesis builds on the idea that refactoring sequences can be generated under explicit objectives, and the quality terms reported by Harman and Tratt, Mohan and Greer, Paixão et al., and Cortellessa et al. inform the weights of the process-quality score (§3.2.2). The difference is in what the generated sequence is used for. In recommendation and navigation systems, the generated sequence is usually the output: it is the path the developer is meant to follow. In this thesis, the generated trajectory is a counterfactual reference. The developer’s observed trace is the object being evaluated, and the synthesised alternative exists only to give that trace something concrete to compare against. Existing recommendation and navigation systems do not judge the intermediate choices the developer actually made, nor explain where the observed path moved away from a better alternative [12, 13]. That is the gap this thesis addresses.

2.5.2 Recent trajectory-formalism work in software engineering

Recent software-engineering work has also begun to describe development in terms of trajectories. RefactorBench, for example, represents agent behaviour as a sequence of actions and observations in a POMDP-shaped environment [9]. Other work looks for recurring action patterns in agent traces [10] or builds training environments around software-engineering tasks [24]. Alomar et al. similarly note that evaluation remains an open issue in LLM-based refactoring research [11]. These papers are useful because they treat the path through a task as something worth analysing, not just the final answer. However, their focus is usually on language-model agents and benchmark success, rather than on the quality of a human developer’s refactoring process. Observational work on real refactoring practice is closer to the setting of this thesis: it shows that refactoring normally unfolds as an ordered sequence of operations, often mixed with feature work, fixes, or cleanup [25]. This thesis adopts the human-developer scope. Agent sessions appear in Chapter 4 as a feasibility extension applied to the same scoring formula and detector design, not as a second contribution.

2.6 Process-aware work and IDE trace capture

Evaluating refactoring as a process requires more than the before and after versions of the code. It needs a trace: a record of the intermediate states and the actions that produced them. Version-control histories provide one source of this information, but commits and checkpoints are often too coarse. IDE-level logs can capture a finer view of what the developer actually did, including edits, navigation, test runs, and refactoring commands. This section therefore focuses on process-aware logging work, both because it shows that useful traces can be collected and because it exposes the practical difficulties of interpreting them.

2.6.1 Logging developer interactions in IDEs

Several studies have used developer-interaction logs to recover workflows and common behaviour patterns. Damevski et al., for example, mine sequences of Visual Studio interactions from telemetry-like event streams, aiming to find frequent usage patterns without recording the full source code [26]. This is useful evidence that commands, navigation actions, and edits can be captured in a structured form while still protecting developer privacy. The limitation is that such logs do not preserve enough project state to replay a session later, which restricts the kind of process-quality analysis they can support.

Poncin et al. take a broader repository-level view, combining version-control, bug-tracker, and mail-archive data to infer developer workflows using process mining techniques [27]. Their work shows how low-level records can be turned into higher-level activity models, such as sequences of commits, issue updates, and discussions. It also illustrates a recurring problem: development traces are noisy. Developers may work on several tasks at once, and raw events have to be segmented before they can be treated as meaningful steps. Repository-level traces also miss IDE-specific information, such as which command was run or whether a refactoring was invoked through the IDE.

More recent logging infrastructure has often focused on developer–LLM interaction rather than refactoring-specific traces. *CodeWatcher*, for example, is a lightweight VS Code extension that records fine-grained IDE events such as LLM-tool insertions, deletions, copy-paste actions, and focus changes, with timestamps that support later reconstruction of a coding session [28]. The same style of event capture would be useful for refactoring, but the setting does not map directly onto this thesis: the work is centred on VS Code and LLM-assisted coding, whereas this project studies Java refactoring sessions in IntelliJ.

Recent trace-capture work is therefore relevant in principle, but not directly reusable for the setting studied here.

Taken together, existing datasets do not provide exactly the trace needed for this thesis. Some capture rich IDE events, but for a different activity; others capture refactoring-related activity, but only at repository granularity; and others do not retain enough code content to replay the trajectory. The dataset used here therefore has to be built for the task rather than reused directly from prior work. Chapter 3 returns to this in §3.7.

2.6.2 Granularity, segmentation, and semantic interpretation

IDE logging also raises a modelling question: what counts as one “step” in a refactoring process? Commits are usually too coarse, since they often mix refactoring with feature work, fixes, or cleanup. Raw IDE events are too fine-grained: navigation, edits, refactoring-tool calls, and test runs appear as separate events even when they belong to one conceptual change. The process-mining and interaction-analysis literature shows that grouping this kind of activity into meaningful tasks or episodes is difficult [27, 26]. For refactoring, this matters because one refactoring may involve many small interactions, while a short burst of editing may contain several distinct actions.

A process-evaluation system therefore needs a step definition that is both reliable to extract and meaningful to interpret.

2.7 Summary and positioning: toward process-quality evaluation

The literature discussed in §2.1–§2.6 gives a strong starting point for this thesis, but it also shows where existing work stops short of process-level evaluation.

Refactoring itself is well established as behaviour-preserving change, supported by catalogues of operators and by tool-based precondition checking [1, 2, 4]. Most evaluation work, however, still judges refactoring by comparing start and end snapshots, using structural metrics, code-smell counts, or related quality indicators [14, 18]. These signals are useful, but they are imperfect and can disagree across tools, so combining several signals is often more reliable than relying on one measure alone [19, 20, 21]. Refactoring mining and AST differencing make it possible to infer actions from code changes and reconstruct traces from repository histories [7, 8]. Recommendation and synthesis systems show that refactoring sequences can be generated and planned towards final goal states under explicit quality objectives [12, 13, 14, 15], but treat the generated sequence rather than a developer’s observed trace as the object of evaluation. IDE logging work adds that fine-grained traces can be captured, but also shows how difficult they can be to segment and interpret [27, 26, 29].

There has therefore been substantial work on *detecting* refactorings, *recommending* refactorings, and *synthesising* refactoring sequences. What is still missing is a system that scores the path a developer actually took against an explicit process-quality metric, then compares that path with higher-scoring alternatives. The closest existing work usually treats the generated sequence as the answer: if it reaches a good target state, the system has done its job. In this thesis, the observed developer trace is the object being evaluated. The aim is to judge whether the process itself was good, including whether it introduced unnecessary intermediate degradation, disruption, or instability, and to explain where it diverged from better-scoring reference trajectories.

This thesis takes that missing comparison as its starting point. For a recorded refactoring session, it scores the developer’s trace and builds a higher-scoring reference path between the same start and end states, so the two can be compared directly. This leads to

the overall thesis question: *given the start and end state of a refactoring session, can we judge the quality of the path the developer took and point to places where a better path was available?*

Chapter 3

Methodology

This chapter explains how the proposed process-evaluation approach is defined, implemented, and evaluated. It begins by setting out the state/action/trace notation used throughout the chapter (§3.1). It then defines the process-quality score (§3.2.1–§3.2.3) and describes the event-capture, shadow-repository, and metrics infrastructure that supplies the data for that score (§3.2.4–§3.2.6). Next, it introduces the divergence-point detector (§3.3), the Eclipse JDT substrate used to construct alternatives (§3.3.1), the per-kind alternative trajectory synthesisers (§3.3.2–§3.3.5), and the validator used to check that each alternative still reaches the same end state (§3.3.6). The chapter then describes the overall system architecture and offline/replay pipeline (§3.4.1–§3.4.2), followed by the participant dashboard (§3.5). Finally, it defines the evaluation methodology used in Chapter 4: the score-formula sensitivity and ablation design (§3.6), the 45-session dataset and labelling protocol (§3.7), and the divergence-detector evaluation protocol (§3.7.6).

3.1 Formal definitions and notation

The formal definitions in this section build on the state/action/trace framing from §2.1. A refactoring trajectory is a sequence

$$\tau = (s_0, a_0, s_1, a_1, \dots, a_{T-1}, s_T),$$

where each s_t is a code snapshot and each a_t is a developer action that turns s_t into s_{t+1} . Actions are either IDE-supported refactoring operations or manual edits; in practice a single action can carry both a structured refactoring spec (recovered by RefactoringMiner [7]) and some leftover unstructured edits. We treat this as one action with two parts rather than two separate actions.

The tool evaluates a trajectory by producing an *analysis report*. For a trajectory τ , this report is written as the tuple $(\tau, \mathcal{M}, \mathcal{D}, \mathcal{A})$, where:

- $\mathcal{M} = \{m_0, m_1, \dots, m_T\}$ is the set of metrics collected for the states in the trajectory. Each m_t contains the cleanliness sub-score $C(s_t)$ and the raw signals used to compute it, including PMD violations, CPD duplications, Chidamber–Kemerer measures, build/test outcomes, the action’s refactoring spec, and commit and test event metadata.
- $\mathcal{D}$ is the set of divergence points the detector emits for τ (§3.3).
- $\mathcal{A}$ is the set of alternative reference trajectories synthesised at those divergence points (§3.3.2 onwards). Each $\tau^* \in \mathcal{A}$ is represented in the same way as the observed trajectory, which allows both trajectories to be scored using the same process-quality score.

The process-quality score $J(\tau)$ is computed over the full trajectory using $\mathcal{M}$, and is the focus of §3.2.1–§3.2.2. Table 3.1 summarises the notation used throughout the chapter.

Symbol	Meaning
$\tau = (s_0, a_0, s_1, \dots, a_{T-1}, s_T)$	Refactoring trajectory
s_t	State (code snapshot) at step t
a_t	Action at step t (IDE refactor or manual edit)
$J(\tau)$	Process-quality score, clamped to $[0, 100]$
$C(s)$	Cleanliness sub-score of state s
$\Delta C(\tau) = C(s_T) - C(s_0)$	Cleanliness gain across trajectory
$\beta = 50$	Baseline anchor in the process score
W_x	Process-score weight for term x
r_b, r_{st}, r_{mi}	Laplace-smoothed rate terms (broken, skip-tests, manual-when-IDE)
$n_{cg}(\tau)$	Commit-gap event count
$\sigma_l(\tau)$	Step-savings bonus for alternative trajectories
DP	Divergence point
τ^*	Alternative reference trajectory
$\mathcal{K}$	Divergence-kind set = $\{\textit{Ordering}, \textit{Manual-Refactor}, \textit{Rework}, \textit{Hygiene}\}$

Table 3.1: Notation used throughout the methodology chapter.

3.2 Computing the process score

This section defines the process score $J(\tau)$ and the cleanliness sub-score used inside it (§3.2.1–§3.2.3). It then describes how the tool captures events, reconstructs shadow repositories, and calculates the per-checkpoint metrics needed to compute the score (§3.2.4–§3.2.6).

3.2.1 The process score

The process score $J(\tau)$ combines seven weighted terms across the trajectory and clamps the result to a $[0, 100]$ range so trajectories of different lengths can be compared on the same scale:

$$J(\tau) = \text{clip}_{[0,100]} \left(\beta + W_G \Delta C(\tau) - W_B r_b(\tau) - W_{ST} r_{st}(\tau) - W_{MI} r_{mi}(\tau) - W_{CG} n_{cg}(\tau) - W_{LAG} \ell(\tau) + W_L \sigma_l(\tau) \right) \quad (3.1)$$

where:

- $\beta = 50$ is the baseline. A trajectory with no cleanliness gain and no process penalties scores 50.
- $\Delta C(\tau) = C(s_T) - C(s_0)$ is the cleanliness gain from start to end (§3.2.3).
- $r_b(\tau) = b_{ms}(\tau)/e_{ms}(\tau)$ is the fraction of wall-clock time for which the build or test suite was broken.
- $r_{st}(\tau)$ and $r_{mi}(\tau)$ are event-rate penalties. They use Laplace smoothing, $r_\bullet(\tau) = (k + 1)/(n + 2)$, where k is the number of penalised events and n is the number of opportunities. If no relevant events have occurred yet, the term is set to zero.
- $n_{cg}(\tau)$ counts commit-gap events. One event is recorded for each stretch of at least six green-refactor checkpoints between user commits.
- $\ell(\tau) \in [0, 1]$ measures intermediate-cleanliness lag. It is the per-checkpoint running mean of $\max(0, C(s_T) - C(s_i))$ over the trajectory, in scalar cleanliness units. This

term is positive while the trajectory remains below its final cleanliness level, so it penalises paths that spend longer in degraded intermediate states.

- $\sigma_l(\tau)$ is a step-savings bonus for synthesised alternatives. It contributes only when τ is an alternative trajectory that reaches the same end state in fewer steps than the observed user trajectory.

Three design choices shape this formula. First, the penalty for a broken build or test suite must be larger than any plausible single-step cleanliness gain. This prevents a trajectory from being rewarded overall for improving structure while temporarily breaking correctness. The weight ordering $W_B:W_{ST}:W_{CG} = 28:14:7$ encodes this priority in §3.2.2. Second, Laplace smoothing prevents the event-rate terms from being dominated by one early event on a very short trajectory. Third, clamping the score to $[0, 100]$ makes trajectories easier to compare, but it also introduces saturation: when a user’s trajectory is already close to 100, the improvement available to a higher-scoring alternative is compressed by the ceiling. The sensitivity experiment in §4.1.1 treats this as a known limitation of the score formula.

The score is intended as a consistent and literature-grounded way to compare refactoring trajectories, not as a uniquely correct measure of process quality. There is no held-out ground truth for “true refactoring-trajectory quality” against which the weights could be calibrated directly. The empirical evidence in Chapter 4, including the sensitivity sweep, ablation study, and inter-rater agreement analysis, therefore evaluates robustness, term importance, and classifier accuracy rather than proving that the chosen weights are optimal.

3.2.2 Process-score weights

Table 3.2 lists the seven weights used in Equation (3.1) and summarises the literature behind each one. This section explains why the terms are included. Their empirical effect on the ranking is examined later through the ablation analysis in §4.1.2, which tests all $2^7 = 128$ weight subsets per fixture and reports both solo and leave-one-out recovery against the full formula.

Term	Weight	Cited justification
W_G (gain)	50	Cedrim et al. 2017 [30]; Paixão et al. 2017 [17]
W_B (broken)	28	Paixão et al. 2017 [17]; Gautam et al. 2025 [9] (FM #2)
W_{ST} (skip-tests)	14	Murphy-Hill et al. 2009 [5]
W_{MI} (manual)	11	Negara et al. 2013 [31]; Vakilian et al. 2012 [32]
W_L (length)	11	Harman & Tratt 2007 [14]; Mohan & Greer 2019 [15]
W_{LAG} (lag)	11	Cunningham 1992 [33]; Kruchten et al. 2012 [34]; Cedrim et al. 2017 [30]; Paixão et al. 2017 [17]
W_{CG} (commit-gap)	7	Tao & Kim 2015 [35]; Di Biase et al. 2019 [36]; Herzig & Zeller 2013 [37]; Kudrjavets et al. 2022 [38]

Table 3.2: Weights used in Equation (3.1) and the literature supporting each term. The empirical contribution of each term is examined in the ablation analysis of §4.1.2.

The rest of this section explains each weight in turn. For two terms, W_{MI} and W_{CG} , the literature supports the underlying concern but does not give numerical evidence for how large those penalties should be. Those limitations are noted in the relevant paragraphs.

W_G (**gain, 50**). This is the main positive term in the score: each unit of cleanliness gain at the end of the trajectory adds 50 to $J(\tau)$. Cedrim et al. [30] found that refactoring often leaves structural metrics unchanged or slightly worse, so improvement should be rewarded explicitly rather than assumed. Paixão et al. [17] model the balance between disruption and improvement as something developers actively manage, which motivates treating cleanliness gain and process penalties as competing parts of the same score.

W_B (**broken, 28**). This is the largest single penalty term. Paixão et al. [17] show that developers place substantial value on avoiding disruptive intermediate states, even when doing so means giving up roughly a quarter of the available metric improvement. The penalty is time-weighted rather than binary: $r_b(\tau) = b_{ms}/e_{ms}$, so a brief broken state is penalised less than a long one. The numerator b_{ms} counts only wall-clock time during which the build or tests are failing, rather than total refactoring time.

Time-weighting also makes the term less sensitive to how finely the trajectory is sampled. The state/action/trace framing in §3.1 allows a state to be a commit, an IDE event, or a developer-defined checkpoint. A binary count of broken states would vary substantially across those choices, whereas wall-clock time spent broken is independent of the sampling rate. This choice is also consistent with Gautam et al.’s RefactorBench analysis [9]: their failure-mode #2 shows that rejecting every intermediate broken state can hurt LM-agent performance, because some broken intermediates are necessary during multi-file edits. The weight is therefore set high enough that a plausible single-step cleanliness gain, $W_G \Delta C$, cannot outweigh leaving the build or tests broken.

W_{ST} (**skip-tests, 14**). This term penalises 60-second refactoring windows that finish without a test run. The window size follows Murphy-Hill et al. [5], whose study of real refactoring practice shows that developers tend to run tests after batches of changes rather than after every individual edit. The weight is set to half of W_B : skipping tests is weaker evidence of process damage than an observed broken state, but it still indicates reduced feedback during the refactoring process.

W_{MI} (**manual-when-IDE, 11**). This term penalises steps where the developer performs a refactoring manually even though the IDE could have applied the same transformation automatically. Mens & Tourwé [4] note that IDE refactoring tools apply static precondition checks that manual edits may bypass, making the automated route safer in many cases. At the same time, manual refactoring is common: Negara et al. [31] found that, across 23 developers and 5,371 refactorings, developers often performed refactorings by hand even when an automated equivalent was available. Vakilian et al. [32] also show that automated refactorings can be misused, so the term is treated as a graded penalty rather than a hard rule.

No published study that we are aware of directly measures the fault-rate gap between manual and IDE refactoring, so this weight is not fitted to such data. Instead, it follows the severity ordering set out above. The ablation analysis in §4.1.2 shows that `manualIde` is one of the three largest single-term contributors by sum-over-sum recovery to session-wide divergence-point magnitude, alongside `length` and `broken` (Table 4.5), and the sensitivity sweep in §4.1.1 indicates that the ranking is robust to perturbations of this weight.

W_L (**length, 11**). This is a positive term rather than a penalty. It increases $J(\tau)$ only for alternative trajectories that reach the same end state as the observed user trajectory in fewer steps. Search-based refactoring work treats the number of refactoring operations as an optimisation objective alongside quality [14, 15], and later many-objective work continues this by trading effort against code-quality objectives [23]. The observed user trajectory

is not penalised for length, since a longer path may reflect a larger or more complex task rather than a worse process. The term therefore rewards shorter valid alternatives without treating every long trajectory as poor.

W_{LAG} (lag, 11). This term penalises trajectories that spend time in a worse cleanliness state before reaching their final value. The gain term $W_G \Delta C$ only compares the start and end states, so two trajectories receive the same gain score if they finish with the same cleanliness improvement, even if one cleans up early and the other leaves the code degraded until the end. The lag term adds a cost for this delay.

The motivation is similar to technical debt. Cunningham’s debt metaphor [33] and Kruchten et al.’s later formalisation [34] both treat degraded code as something that becomes more costly the longer it remains. Cedrim et al. [30] give empirical support for applying this idea to refactoring sequences: they show that intermediate states can degrade structural metrics relative to the start, even when the final result is an improvement. The lag term therefore penalises time spent below the trajectory’s final cleanliness target, rather than only measuring the endpoint gap.

The weight is set to 11, the same as W_L and W_{MI} , and half of the broken-state penalty W_B , following the severity ordering in §3.2.2. Since $\ell \in [0, 1]$, the maximum lag penalty is 11 points, which keeps it smaller than the maximum gain credit of 50. The ablation results give the strongest support for this term in the *Ordering* category: §4.1.2 reports its solo magnitude-recovery, and §4.2.2 shows that the number of *Ordering* divergence-points where the synthesised alternative beats the user’s trajectory rises from 4 to 10 of 24 once W_{LAG} is included.

W_{CG} (commit-gap, 7). This term penalises long runs of green checkpoints, where the refactoring checks and tests pass, without a commit in between. The aim is to capture a small process cost: when several valid steps are left bundled together, the resulting change is harder to review, explain, or roll back than if the work had been split into smaller confirmed units.

The evidence comes from work on decomposing composite changes. Tao & Kim [35] report that separated commits let reviewers inspect changes more effectively in the same amount of time. Di Biase et al.’s controlled experiment [36] points in the same direction: decomposed changes produced fewer comments that incorrectly marked correct code as defective (1 rather than 6, $p = 0.03$). Tangled commits can also make later defect attribution less reliable [37].

This is deliberately a narrow claim. Kudrjavets et al. [38] find no correlation between PR size and time-to-merge across 1.25M PRs in ten languages, so the term is not meant to capture merge speed. It is instead used as a smaller hygiene signal for review comprehensibility and rollback locality. Because no published study that we are aware of directly links commit cadence during refactoring to these costs, the weight is not fitted to data. The ablation results are consistent with this lower-severity role: `commitGap` recovers non-zero session-set magnitude on its own (sum/sum 0.126), and removing it changes the ranking (mean $\tau = 0.926$) while leaving most total magnitude intact (sum/sum 0.892).

3.2.3 The cleanliness sub-score

The cleanliness sub-score $C(s)$ in Equation (3.1) gives each codebase state a normalised quality value in $[0, 100]$. It is built from six signals rather than one, following the argument in §2.3 that individual quality measures are noisy and context-dependent. Combining several signals makes the score less dependent on any single metric. Table 3.3 lists the six sub-signals, the source used to justify each one, and how each value is computed.

Sub-metric	Weight	Cited basis	Computation source
Cognitive complexity	1.0	Campbell 2018 [39]; Muñoz Barón et al. 2020 [40] (partial validation); Lavazza et al. 2023 [41] (contradicting)	Per-method mean
Coupling (CBO)	1.0	Chidamber & Kemerer 1994 [42]	CK library, per-class mean
Duplication	1.0	Heitlager et al. 2007 [43]; Fowler 1999 [2]	CPD-detected cloned lines on touched files
Readability	1.0	Buse & Weimer 2010 [44] (feature taxonomy only - not the trained model)	Five-feature blend, line-weighted
Smells	1.0	Marinescu 2004 [45]; Arcelli Fontana et al. 2016 [20]; Moha et al. 2010 [18]	PMD violation count on touched files
Cohesion (TCC)	1.0	Bieman & Kang 1995 [46]	Per-class mean, dropped if < 2 methods

Table 3.3: The six sub-signals used to compute the cleanliness sub-score $C(s)$. Each is given weight 1.0; the rationale for using equal weights is given in the composition rule below.

Composition rule and sub-weights. Each raw sub-signal is min-max normalised against the main trajectory’s range and inverted for “lower is better” measures (cognitive complexity, coupling, duplication, smells). Sub-signals with degenerate ranges or missing data on the trajectory are dropped and the remaining weights are renormalised. Alternative-trajectory checkpoint values are clamped to $[0, 1]$ against the main trajectory’s range, so an extreme alternative cannot shift the main trajectory’s normalisation. The final cleanliness score is then computed as a weighted sum and scaled to $[0, 100]$.

All six sub-weights are set to 1.0. Weighted sums are widely used in quality models such as Quamoco [47] and QMOOD [48], but those models do not tell us how these particular signals should be balanced across a refactoring trajectory. Existing calibrated composites, such as Coleman et al.’s Maintainability Index [49], are fitted to absolute metric values on single snapshots, not to normalised changes across checkpoints. They therefore do not transfer directly to this setting. In the absence of trajectory-level calibration data, we use equal weights as the least assumptive choice. The sensitivity experiment in §4.1.1 supports this decision: perturbing the cleanliness sub-weights changes only a small fraction of divergence-point rankings, an order of magnitude fewer than the process-score perturbations.

Caveats. Two of the sub-signals have weaker empirical backing than the others:

- *Readability.* We use Buse & Weimer’s feature taxonomy [44] (line length, indentation, identifier characteristics) as a feature blend, but not their trained readability model. Calibrating the trained model for trajectory evaluation would require domain-specific data that we have not collected. For this reason, readability is used only as one component of the broader cleanliness score.
- *Cognitive complexity.* The evidence for cognitive complexity is mixed. Muñoz Barón et al. [40] re-analysed 24,400 human understandability ratings and found positive correlations with comprehension time and subjective ratings, although the link with correctness was weak. Lavazza et al. [41], by contrast, found that cognitive complexity predicts understandability about as well as traditional size and cyclomatic-complexity measures, with little extra explanatory value. We therefore treat it as a useful but limited signal within the six-part score.

Coupling to the process score. The process score uses the cleanliness sub-score in two places. First, it uses the endpoint change $\Delta C(\tau) = C(s_T) - C(s_0)$. Second, it uses the final cleanliness value $C(s_T)$ as the reference point for the lag term $\ell(\tau)$. This means that changing a cleanliness sub-weight can affect both the endpoint reward and the lag penalty. This is intentional: the lag term only makes sense if it is measured in the same cleanliness units as the rest of the score. It also has a visible effect in the sensitivity analysis. On the agent sessions, perturbing cleanliness weights by $\times 0.5 / \times 1.5$ lowers Kendall’s τ by 3–10 % relative to the pre-lag formula (§4.1.1). The injection and user-study sessions do not show measurable rank disruption from the same perturbations.

The process score and cleanliness sub-score are computed from the state of the codebase at each checkpoint in a recorded session. The next three subsections describe how those checkpoint states are produced and measured: event capture in the IDE plugin (§3.2.4), shadow-repo reconstruction that materialises the codebase at each checkpoint (§3.2.5), and per-checkpoint metrics calculation (§3.2.6).

3.2.4 Event capture

The IntelliJ plugin records the events emitted during a user’s refactoring session. These include editor events (files opened, closed, saved, modified, renamed, and moved), refactoring events (refactoring start and finish), build events, test-run events with pass/fail counts, git events for newly observed commits, and session start/end events.

File modification activity is the noisiest source. IntelliJ emits an event for every keystroke, so we debounce these events before storing them. The debouncing layer groups keystrokes into an *edit burst* across all files: once there has been no further typing for two seconds, it emits a single burst event containing the file’s post-edit contents and the structured set of program elements touched.

Some fine-grained file-level events, such as file-open and file-save events, are not used by the current analysis. We still record them because they are cheap to capture in the plugin and may be useful for later work on navigation patterns, file engagement, and read-versus-edit behaviour.

Before downstream processing, the captured events are re-sorted by timestamp and spurious events are removed. The resulting *normalised event stream* is consumed by every subsequent stage.

3.2.5 Shadow repo reconstruction and worktree pool

Analysing a user’s trajectory requires the ability to reconstruct, and cheaply check out, the exact state of the codebase at any point during the session. We do this by replaying the normalised event stream into a fresh per-session git repository, with one commit per state-bearing event. The baseline commit is the initial source snapshot taken at session start; each later event that carries file changes applies its snapshots to the working tree, stages them, and commits. No-op events – those that carry no file changes, or whose staged diff is blank-line-only – map to the previous SHA without producing an empty commit. The output is a per-event map from event id to SHA, which gives every downstream stage a stable (from-SHA, to-SHA) pair to check out.

3.2.6 Metrics calculation

For every unique state in the shadow repo, the pipeline computes a checkpoint metric record. This record contains the build and test outcome, the cleanliness signals (PMD violations, CPD duplications, code readability, and Chidamber–Kemerer structural measures), and the file-level diff against the previous checkpoint.

Build and test outcomes are produced by checking the project state out into a worktree and running the project’s own Gradle build and test tasks. For each checkpoint, the pipeline records whether the build succeeded and whether the test suite passed. These outcomes are computed for every checkpoint, regardless of whether the user ran the build or tests during the original session.

This stage dominates the wall-clock cost of the pipeline, so the implementation reuses as much build state as possible. All checkpoints share a single persistent worktree, which is the only stage that bypasses the fresh-per-borrow rule of §3.2.5. The pipeline then walks the unique SHAs in order using detached checkouts. Because the worktree’s build directory is preserved across SHAs, the Gradle daemon, Gradle build cache, and incremental compiler can all reuse prior state at the next checkpoint.

The static-analysis signals are tracked across checkpoints rather than only computed independently at each checkpoint. Each PMD violation and CPD clone group is threaded through the unified diffs, allowing the report to identify when a violation was introduced, when it was resolved, and which transition was responsible. This is what lets the dashboard show, for any refactoring step, the smells it introduced or eliminated. Two user-study participants found this inspection feature valuable during interviews (§4.3.1).

3.3 Divergence points

A *divergence point* is a step where the user’s trajectory can be compared with an alternative path that scores strictly higher under the process score. The detector reports four kinds of divergence:

$$\mathcal{K} = \{ \textit{Ordering}, \textit{Manual-Refactor}, \textit{Rework}, \textit{Hygiene} \}.$$

Each reported point records the divergence *kind*, the *anchor step* where it occurs, and its *magnitude*. It also stores the context needed to explain that kind of divergence: the reorder window for *Ordering*; the replaced refactoring identifier for *Manual-Refactor*; the originating-terminal step pair, file, scope, and reverted-line count for *Rework*; and the hygiene sub-kind and stretch length for *Hygiene*.

The magnitude is uniform across all four kinds:

$$\text{magnitude} = J(\tau_{\text{final}}^*) - J(\tau_{\text{final}}),$$

where τ_{final}^* is the alternative path substituted into the user’s trajectory at the divergence point, with the rest of the user’s later activity left unchanged. τ_{final} is the score of the user’s actual trajectory. Both scores are therefore compared at the same final point, which makes the magnitude comparable across the four divergence kinds: “*if you had taken this one different path at the divergence point and continued with the rest of your trajectory unchanged, by how many process-score points would your final score have moved?*”

Before emitting a divergence point, the detector applies filters specific to each kind. Three thresholds are used. $\text{min_rework_lines} = 2$ removes single-line *Rework* noise. $\text{min_commit_gap} = 6$ requires at least six green-refactor checkpoints between commits before *Commit-Gap* is reported. The 60s composite-window boundary [5] defines the refactoring batch boundary used by both *Ordering* and *Hygiene*. The value 6 is based on Murphy-Hill’s batching heuristic rather than on a measured cutoff from prior work. Since the 60s window already groups tightly clustered refactorings, requiring six consecutive green checkpoints helps distinguish a real no-commit gap from normal in-batch activity. No published study calibrates this exact threshold, so it is treated as a tunable parameter. The sensitivity analysis in §4.1.1 tests the associated weight W_{CG} , not the threshold itself.

The remainder of the section explains each divergence kind together with the synthesiser that constructs its alternative trajectory. §3.3.1 introduces the Eclipse JDT support used

for *Ordering* and *Manual-Refactor*. §3.3.2–§3.3.5 then cover the four divergence kinds, followed by the bracket-level validator for reordered trajectories in §3.3.6.

3.3.1 Applying refactorings via Eclipse JDT

Synthesising counterfactual trajectories requires the system to apply refactorings, not just describe them. Rather than building a refactoring engine from scratch, the implementation uses Eclipse JDT. This choice was primarily practical: JDT can be embedded as a library and driven programmatically from a JVM, whereas using IntelliJ’s refactoring engine would require a custom plugin running inside a full IDE instance, adding UI dependencies, process boundaries, and startup overhead. JDT also provides the refactoring operations needed for this analysis through the same mature implementation used inside Eclipse.

Although JDT was designed for use inside Eclipse, it can also be embedded in another JVM by hosting the Equinox OSGi framework. This is necessary because JDT refactorings depend on Eclipse’s workspace model, not just on the Java parser or AST library. The simpler bare-classpath setup is enough for parsing and AST analysis, but not for applying refactorings [50].

Batch sessions. JDT maintains an indexed project model in memory. Building this model takes roughly a second, so repeatedly rebuilding it between refactorings would be expensive. Instead, the pipeline wraps a whole refactoring window in a single *batch session*: the model is initialized once at the start, reused across all operations, and torn down at the end. Each refactoring then sees a fully indexed model without paying the initialization cost repeatedly.

Anchor resolution. Refactorings need to identify the code element they should act on, but raw line numbers are fragile because they change when nearby code is edited. The pipeline therefore uses content-addressed anchors. RefactoringMiner’s line and column coordinates are used only once, on the host side, to find the relevant AST node. That node is then represented by more stable information: the fully qualified name of the enclosing type, the host method name and parameter types, and a canonical hash of the node’s own AST subtree. This anchor is what the host sends to the JDT bundle. For range-based operations, the original line and column are not sent at all; for point-based operations, they are kept only as a tiebreaker in the rare case where two declarations inside the same method have the same subtree hash. The JDT bundle first finds the host method by name and parameter types inside the named type, then searches within it for a node with the matching subtree hash. Because both sides compute the hash in the same way, the lookup is robust to formatting changes and to line shifts elsewhere in the file. For example, a method that moves from line 42 to line 47 still resolves correctly, as long as the method’s own subtree has not been rewritten.

3.3.2 Ordering

Ordering detection is applied to bracket-validated windows of consecutive refactoring steps (§3.3.6). For each window, the reorder synthesiser computes the valid permutations of those steps and measures how each ordering changes the process score relative to the developer’s original order. The system emits at most one divergence point for the window, placed at its final step. Its magnitude is the best score improvement found among the admitted permutations, and the admitted permutations are attached as alternatives.

The reorder synthesiser produces these alternatives by reshuffling the user’s refactoring steps while still requiring the final code state to match the user’s. The following paragraphs

describe how candidate reorderings are generated, filtered, and checked before they are scored.

Window splitting on validator verdicts. Only bracket-validated windows (§3.3.6) are eligible for reordering. A bracket is rejected if the synthesiser cannot reproduce it. This can happen in three ways: *untyped* means that RefactoringMiner could not map the user’s edit to a typed JDT refactoring spec, which is common for unstructured manual edits; *refactor_failed* means that JDT tried to apply the spec but failed; and *ast_diverged* means that JDT applied the spec, but the resulting AST did not match the user’s. If any of these verdicts occurs, the containing bracket is split into individual steps that are not reordered. The wrap-and-patch layer (§3.3.3) handles the most common JDT-IntelliJ differences before this validation step, increasing the number of brackets that pass.

Dependency-DAG construction. Each refactoring step carries a structured *spec* describing what entities it reads, writes, creates, or deletes. The synthesiser builds a dependency graph where an edge from step u to step v means v depends on u ’s effects. This graph defines the constraints on any valid reordering.

SSA versioning across renames. Renames make dependency tracking harder because the same program entity can appear under different names at different points in the trajectory. For example, if a method is renamed from `foo` to `bar`, later steps refer to `bar`, even though it is the same method. The synthesiser handles this by assigning a new version identifier after each rename and carrying that identifier through the dependency graph. Later steps that target the renamed entity are then linked to the correct version. This is needed for cases such as Extract Method followed by Rename Method on the extracted method.

Topological enumeration. The synthesiser enumerates valid topological orderings of the dependency graph using recursive backtracking. Each ordering is one candidate permutation of the window’s refactoring steps. Because the number of possible orderings can grow very quickly, the search has a hard size limit: windows with more than 7 refactoring steps are skipped, and enumeration within an accepted window stops after at most $7! = 5040$ permutations. In the recorded sessions used in Chapter 4, reorder windows rarely reached this limit. However, sessions with longer uninterrupted refactoring sequences could have their largest windows skipped by the *Ordering* detector; this limitation is discussed further in §5.6.

Depth-first search over a prefix trie. The enumeration is implemented as a depth-first search over a prefix trie of candidate permutations. This lets the synthesiser reuse work across orderings that begin with the same sequence of refactorings. When the search moves forward, it applies one refactoring to the worktree; when it backtracks, it rolls the worktree back to the previous state. As a result, two orderings that share their first j steps also share the work needed to materialise those j steps, instead of applying the same prefix repeatedly from scratch.

Terminal AST-equivalence audit. For each full permutation, the synthesiser checks that the final code state matches the user’s final code state. This is the correctness check for reordering: a candidate ordering is only kept if it reaches the same end point as the original trajectory. At each leaf of the trie, meaning the terminal state of one complete ordering, the synthesiser hashes the ASTs of the touched files into a canonical form. These hashes are compared with the user’s terminal AST hashes, which are precomputed from

the user’s own trajectory with `git show` at the start of the window, so a second worktree is not needed. The canonical form ignores differences that should not affect behaviour, such as whitespace, comment position, formatting, and method-declaration order within a class. The method-order case matters because applying an Extract Method earlier or later in the bracket can place the new method at a different lexical position even when the resulting class is otherwise the same. If the terminal hashes match the user’s, the ordering is kept and scored under Equation (3.1); otherwise it is discarded.

Three implementation choices keep this search practical. First, the whole window runs against a single borrowed worktree. Forward steps, backtracking, and the final AST check therefore reuse the same checkout instead of creating a fresh worktree for every candidate ordering. Second, the window runs inside a single JDT batch session (§3.3.1), so each refactoring can reuse the already-initialized project model. Third, when the search backtracks, the worktree is rolled back with a detached git checkout and JDT is asked to refresh its view of the changed files, just as it would after external file changes in an IDE.

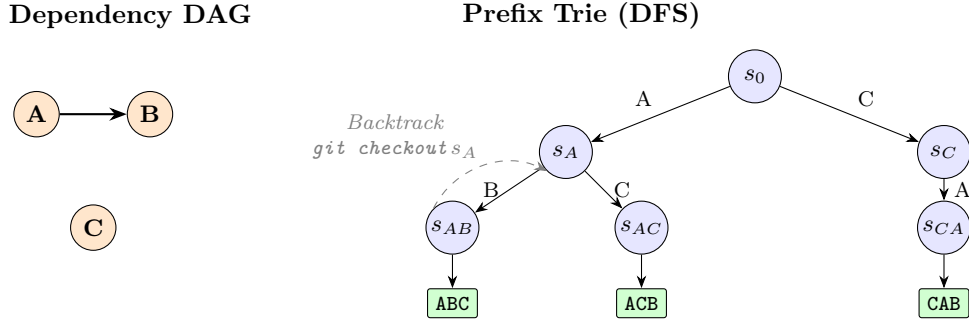


Figure 3.1: Reorder synthesis for a three-step window with one dependency edge ($A \rightarrow B$). The dependency graph on the left permits three valid orderings. The prefix trie on the right shows how shared prefixes are reused: the prefix A is applied once, then reused by both orderings that begin with A .

3.3.3 Manual-Refactor

A *Manual-Refactor* divergence point is emitted at any user step where (i) the action was performed by hand, and (ii) the resulting diff matches the structural pattern of an IDE-supported refactoring. IDE-driven refactorings arrive pre-tagged in the captured event stream, but a manual refactoring is just a series of edit bursts whose net effect happens to constitute a structured refactoring. The pipeline recovers them by running Refactoring-Miner [7] – a standard tool that detects refactorings between two code states by AST-level structural matching – over a sliding window of commit pairs in the shadow repo. The window matters because a manual refactoring may unfold across several intermediate edit bursts, each of which on its own looks like an unrelated change; only by widening the window to cover the full span does the refactoring become visible to the miner. Each detected refactoring is then cross-checked against the captured event stream: a detection whose window contains a matching IDE-emitted refactoring event is recorded as IDE-driven, and one whose window contains no such event is flagged as manual. Adjacent candidate steps that share the same (`fromSha`, `toSha`) bracket are grouped into a single composite that the synthesiser treats as one refactoring.

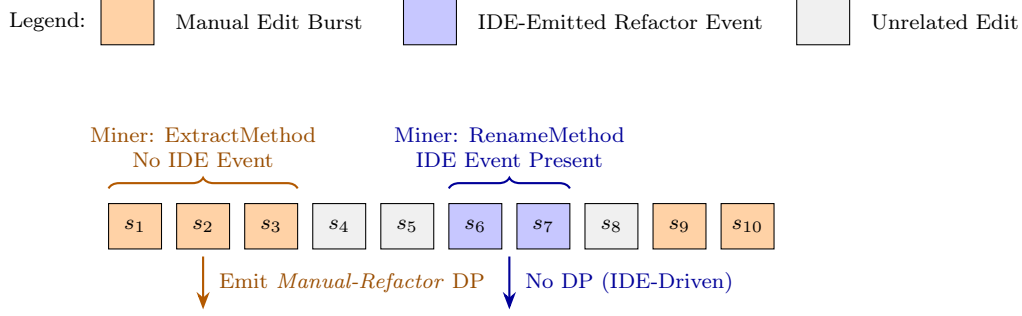


Figure 3.2: Manual-Refactor detection over a sliding edit window. RefactoringMiner detects structured refactorings between bracket endpoints, and each detection is checked against the IDE event stream. A detection with no matching IDE event is treated as manual and emits a divergence point; a detection with a matching IDE event is recorded as IDE-driven.

The manual-refactor synthesiser applies each detected refactoring spec through the corresponding IDE-style operation in JDT. It then runs a residual three-way merge to add back the user’s other edits, such as bug fixes or small unrelated changes that were interleaved with the refactor. The resulting tree is committed to the shadow repository. This merge step is needed so that the alternative trajectory keeps the user’s non-refactoring edits and still ends at the same code state as the original trajectory. The comparison then measures the difference between doing the refactoring by hand and doing it through the IDE, rather than being distorted by missing edits in the final state. Without this merge, the alternative trajectory would diverge from the user’s final code and the process-score comparison would not be meaningful.

The wrap-and-patch layer. One practical issue is that sessions are recorded in IntelliJ, while the pipeline replays refactorings through JDT. The two engines can produce slightly different ASTs even when they are carrying out the same refactoring. To reduce these false divergences, the pipeline includes a wrap-and-patch layer between the JDT output and the validator. During testing, two recurring differences were observed. First, JDT adds the `static` modifier to an extracted method only when the method body requires it, whereas IntelliJ adds it whenever the body permits it. Second, JDT sometimes under-detects outer-scope-variable liveness in conditional branches, producing a void-returning method where IntelliJ produces a value-returning one. A host-side post-pass detects these patterns and rewrites the JDT output to match IntelliJ’s structure. If a remaining difference cannot be patched safely, the bracket is marked as diverged and is not reordered. The wrap-and-patch layer therefore increases the number of usable brackets, but it does not cover every possible JDT-IntelliJ difference; full coverage is left as future work and discussed further in §3.3.6.

3.3.4 Rework

Rework detection looks for edits that are later undone. In practice, this means finding content that is added and then removed, or removed and then added back, so that the two changes cancel out apart from whitespace. The detector parses each step’s unified diff into hunks, maps each hunk to its enclosing method or class, and hashes the affected lines after normalising whitespace. It then greedily pairs added and removed hunks that share the same (file, scope, content-hash). Each matched (originating step, terminal step) pair becomes a *Rework* candidate, while pairs below the `min_rework_lines` threshold are discarded as line-level noise.

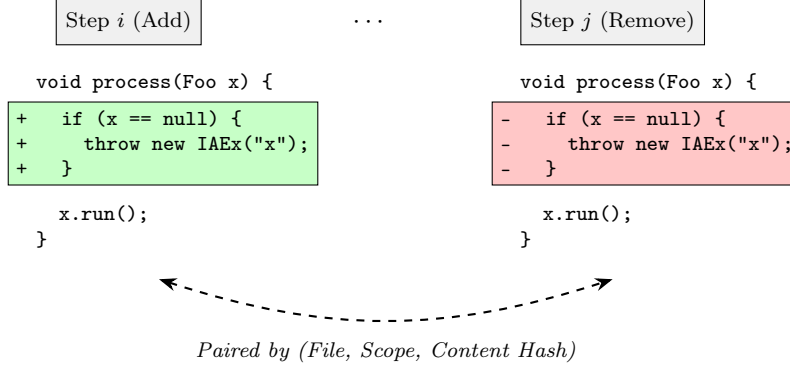


Figure 3.3: Rework detection for an added chunk that is later removed. The detector pairs the two hunks using their (file, scope, content-hash) tuple. The synthesiser then builds a shorter alternative trajectory by removing both halves of the rework and preserving the other intermediate edits.

The rework synthesiser builds an alternative trajectory that skips the matched originating-terminal hunk pair. For each pair, it replays the [originating, terminal] window as a sequence of small patches, leaving out the reworked chunk while preserving the other changes in the window. The replay is treated atomically: if any patch fails to apply, the whole alternative is discarded, so the system never scores a partially reconstructed trajectory.

A key issue is that the intermediate edits have shifted line numbers around the chunk. The terminal-step patch therefore has to translate its line coordinates from the user’s tree, where the chunk reappears, to the synthesised tree, where the chunk was never written and the file may be several lines shorter. To do this, the synthesiser keeps a per-file drift tracker. It records the line offset between the two trees as a set of zones, and each intermediate user hunk shifts the zones after it by that hunk’s net line change. This keeps each zone tied to the same logical content as the user’s tree grows or shrinks around it.

After the reworked chunk is removed, some intermediate checkpoints no longer contain any substantive change: their only edit was the content that has now been omitted. Others contain only whitespace changes, such as blank-line edits around the removed chunk. Empty checkpoints are dropped from the alternative trajectory instead of being committed as no-op steps, while checkpoints whose remaining change is purely whitespace are folded into the preceding checkpoint. Both cases shorten the alternative’s step count, which improves its process score through the step-savings term defined in §3.2.1.

3.3.5 Hygiene

Hygiene detection covers two cases where the refactoring process remains technically valid but lacks a useful safety checkpoint. *Tests-Skipped* is reported once for each 60-second composite window [5] that contains at least one refactoring step but no successful test run. *Commit-Gap* is reported once for each stretch of at least six green-refactor checkpoints, meaning checkpoints where the tests passed and the refactoring applied cleanly, without an intervening user commit. Each finding emits one divergence point.

The hygiene synthesiser builds alternatives in which the user ran tests or made a commit at the relevant checkpoint, without changing the code. For *Tests-Skipped*, the alternative adds a synthetic `TEST_RUN_FINISHED` event at the anchor checkpoint. It does not change the recorded test outcome: `tests.success` and `tests.wasSkipped` remain as they were in the user’s trace. The counterfactual is therefore “the user should have run the tests here”, not “the tests would have passed here”. This affects W_{ST} , which depends on whether tests were run, but leaves W_B unchanged because that term depends on the broken/passing outcome. For *Commit-Gap*, the alternative adds a commit at the anchor checkpoint,

breaking what would otherwise have been a long stretch of uncommitted green-refactor checkpoints.

In both cases, the alternative changes only the process metadata at the anchor checkpoint, not the code. This is because hygiene concerns the process around the refactoring: when tests were run and when commits were made. Synthesis therefore amounts to flipping the relevant process flag at the anchor checkpoint and feeding the trajectory back through Phase B (§3.4). The difference between the original and adjusted final scores gives the process-score cost of the skipped hygiene event.

3.3.6 Validator and behaviour preservation

The bracket-level validator checks whether the user’s specs can be replayed in order to reach the same final state as the user’s trajectory. For each (`fromSha`, `toSha`) bracket, it replays the specs and then applies two checks. First, the set of changed `.java` files must match `git diff fromSha..toSha`. Second, the terminal AST hash for each changed file must match the user’s version. Brackets that pass both checks are marked as *valid* and can be used as reorder windows. Brackets that fail are split into individual steps and are not reordered. The replay uses the same borrowed worktree and JDT batch-session machinery as the reorder engine (§3.3.2).

The number of brackets that pass validation is limited by differences between JDT and IntelliJ, as discussed in §3.3.3. If the wrap-and-patch layer covers the difference, the bracket can still be marked as valid. If it does not, the bracket fails validation even when the original refactoring was correct. Covering all structural differences between JDT and IntelliJ is left as future work, and this limitation is the main reason *Ordering* recall is not higher in §4.2.2.

3.4 System architecture and pipeline phases

This section describes the four JVM modules that make up the tool and explains how they fit into the analysis pipeline. §3.4.1 gives the module-level overview and dataflow, while §3.4.2 explains the Phase A / Phase B split used by the implementation.

3.4.1 Architecture at a glance

The tool is split into four JVM modules (`:ide-plugin`, `:analysis`, `:refactoring-bundle`, and `:shared`) and a separate React/TypeScript dashboard (`dashboard/`):

- `ide-plugin/` – an IntelliJ plugin that records the developer’s session as a stream of typed events and uploads it to the analysis server. The user starts and stops recording with a button in the IDE.
- `analysis/` – the Kotlin analysis pipeline. It ingests recorded sessions, reconstructs a shadow git repository, mines and synthesises trajectories, computes per-checkpoint metrics, and emits an `AnalysisReport`.
- `refactoring-bundle/` – a headless Eclipse JDT distribution wrapped in an embedded Equinox OSGi framework. The analysis module calls into it across the OSGi classloader boundary (details in §3.3.3).
- `dashboard/` – a React/TypeScript single-page app that renders the `AnalysisReport` (details in §3.5).

Figure 3.4 summarises the dataflow between these components.

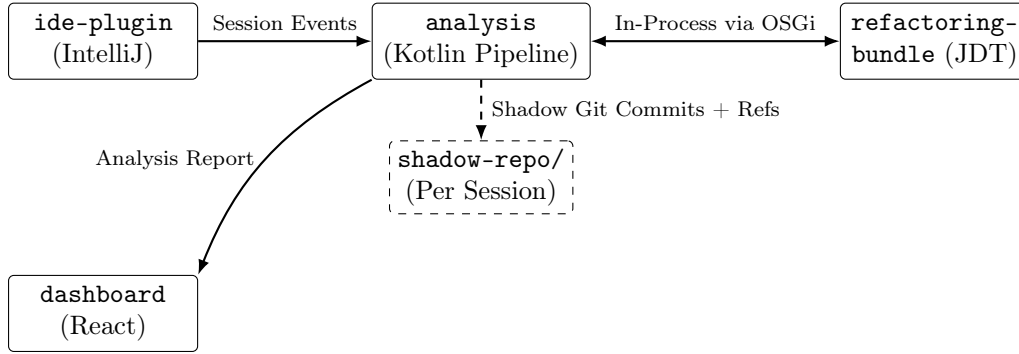


Figure 3.4: High-level dataflow through the tool. The IDE plugin records the session, the analysis pipeline reconstructs and scores the trace, and the embedded JDT bundle applies synthesised refactorings. The dashboard displays the resulting report, while the shadow repository stores user checkpoints and synthesised alternatives as per-session git commits.

3.4.2 Phase A / Phase B split

The detector and synthesisers run as part of an offline analysis pipeline split into two phases. The first phase performs the expensive work: it ingests the recorded events, reconstructs the shadow repository, mines refactorings, runs the synthesisers and validator, computes per-checkpoint metrics, and tracks PMD / CPD findings across checkpoints. This phase usually takes minutes per session, mainly because it runs builds, tests, and JDT refactorings.

Phase A. The output of the first phase is a cached analysis bundle containing the reconstructed trace, mined refactoring steps, per-checkpoint metrics, synthesised alternatives, and supporting diff/quality data. It is saved so that the expensive reconstruction work does not need to be repeated.

Phase B. The second phase takes this cached bundle and a chosen set of score weights. It computes the process score, attaches alternative-vs-user deltas, and produces the final report. Unlike the first phase, it does not reconstruct repositories, apply refactorings, or rerun builds and tests; it only re-scores the cached analysis data, so it finishes in milliseconds.

This split is important for the sensitivity experiments in Chapter 4. Changing the score weights only affects the second phase, so thousands of scoring configurations can be evaluated against the same cached first-phase output without rebuilding trajectories or rerunning tests.

3.5 Dashboard

The detector and synthesisers produce an analysis report, and the dashboard presents that report to participants. The dashboard is a React web app hosted inside the IDE through JCEF, IntelliJ’s embedded Chromium browser, in a dialog rather than a separate browser page, and the plugin passes the completed report to the page once analysis has finished. Figure 3.5 shows the dashboard in use on one of the user-study sessions.

The main view is a trajectory chart showing the user’s process score over time, with synthesised alternatives overlaid for comparison. Participants can toggle the individual code-cleanliness signals on the chart, such as readability, complexity, duplication, code smells, coupling, and cohesion, to see how different aspects of the code changed across the

session. Selecting a step opens a detail panel for the exact edit made at that point, including the affected code, the metrics that changed, and any divergence point associated with the step. Divergence points are clickable: they explain what went wrong, what alternative action the developer *could* have taken, and how much that alternative would have changed the process score. The dashboard also surfaces the locations of code smells and duplicated code, and shows build and test status over time so participants can see where the session became broken and where it recovered. It also highlights the highest-impact and actionable takeaways from the session as a whole.

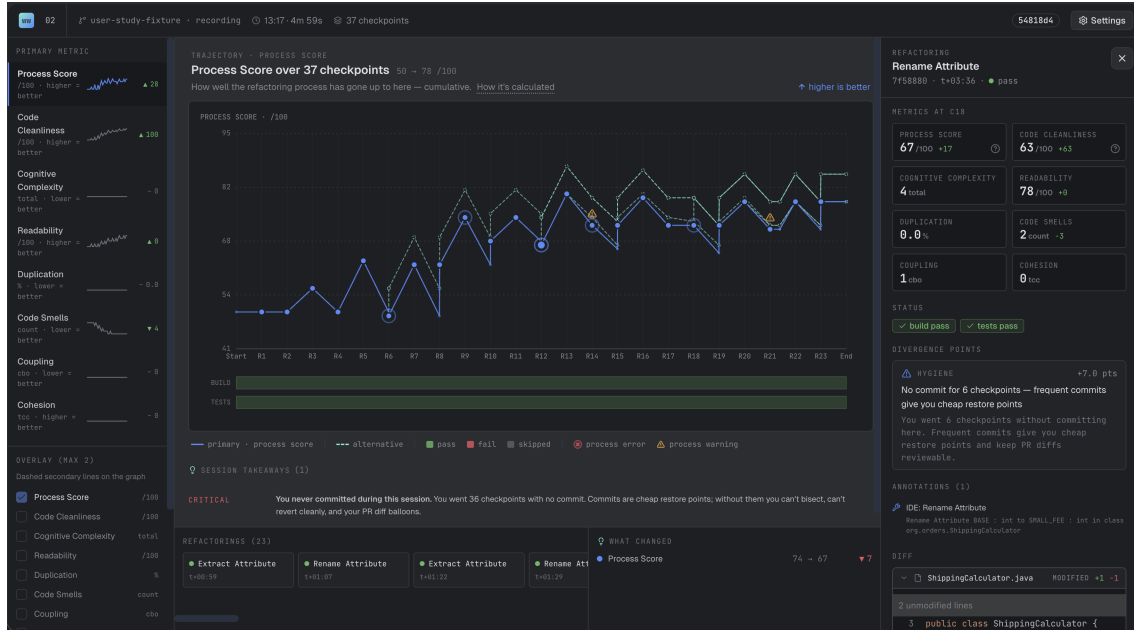


Figure 3.5: The dashboard hosted inside an IntelliJ JCEF window. The centre chart plots the user’s process score across the session and can overlay alternative trajectories and individual cleanliness signals. The side panels show the selected step, metric changes, build/test status, smell and duplication locations, and any divergence point associated with that step, including the suggested alternative and its score impact. The lower panel summarises the highest-impact takeaways for the session.

3.6 Score-formula evaluation design

The results chapter evaluates the score formula in two ways. The first asks whether the ranking of divergence points is stable when the weights are perturbed. The second asks whether the individual terms in the formula actually contribute to the magnitudes the detector reports. This section defines the shared experimental procedure for those tests; Chapter 4 reports the observed numbers.

3.6.1 Sensitivity perturbations

The sensitivity experiment compares the ranking of divergence points computed by the production weights with rankings produced under perturbed weights. Two perturbation schemes are used. In the single-knob sweep, one weight is varied while all other weights remain at their production values. In the multi-knob sweep, all weights are varied at once by drawing independent log-normal scale factors. For weight W_i , the sampled value is

$$W'_i = W_i \cdot \exp(\sigma Z_i),$$

where $Z_i \sim \mathcal{N}(0, 1)$ and $\sigma = \ln 2$. This makes the perturbation symmetric in log-space: multiplying and dividing by the same factor are treated as equally large changes, and most samples remain within a few multiples of the production value.

Ranking stability is measured with Kendall’s τ -b [51, 52] and top- N hit rate. Kendall’s τ -b is used because the rankings are short and may contain ties. A value of $\tau = 1.0$ means the perturbed ranking is unchanged, while $\tau < 0$ means the order has mostly inverted. Top- N hit rate asks how many of the production top- N divergence points remain in the perturbed top- N . Top-1 is the main comparison; larger N values are only informative for sessions with enough surfaced points. Both metrics are mechanically 1.0 on sessions with ≤ 1 detected divergence point — a single-item ranking cannot be perturbed and a zero-item ranking is undefined — so the rank-based statistics in Chapter 4 are computed on the *rankable subset* of each session set: sessions whose analysis report contains ≥ 2 detected divergence points (any magnitude sign). Magnitude-based statistics (mean recovery, sum-over-sum) continue to use the full session sets, since magnitudes are well-defined on every session. Ties are broken by ascending step index so that production and perturbed rankings are compared deterministically.

3.6.2 Ablation design

The ablation experiment tests whether the score formula depends on each of the seven non-sub-cleanliness terms: gain, broken-build time, skipped tests, manual-when-IDE, length, lag, and commit gap. It enumerates subsets of these process weights, keeps the selected terms at their production values, and sets the others to zero. Since there are seven non-sub-cleanliness terms, the full power set contains $2^7 = 128$ variants, which is small enough to enumerate exactly. This follows the same logic as composite-indicator sensitivity analysis [53] and exact Shapley-style feature attribution when the number of features is small [54, 55].

Two recovery measures are used to describe how much divergence-point magnitude remains under an ablated formula. Mean recovery computes the recovery ratio per fixture and averages those ratios. Sum-over-sum recovery instead sums $|\text{magnitude}|$ across all divergence points in all fixtures and divides by the same sum under the production formula. Sum-over-sum is treated as the main measure because it avoids giving empty or near-empty fixtures the same weight as fixtures with many surfaced divergence points. Both measures are reported on the full injection set (45 sessions) because magnitude is well-defined on every session regardless of per-session DP count; the cross-set rank comparison in Chapter 4 uses the rankable subset of each set, as defined above.

3.7 Evaluation dataset and labelling protocol

The experiments in Chapter 4 use a set of 45 refactoring sessions that we recorded by hand on a Java library-style fixture in IntelliJ. This section explains how the sessions were assembled, how they were labelled, and how the labels were checked by raters.

The IDE-event capture literature surveyed in §2.6 shows that detailed development traces can be collected. However, we did not find an existing dataset that combines per-event code content with IDE-side refactoring events for Java sessions recorded in IntelliJ. This matters for the IDE-versus-manual signal. RefactoringMiner can recover a structural refactoring from a pair of code states, but it cannot tell whether the developer used the IDE refactoring command or typed the change by hand. Since that distinction is part of the manual-when-IDE penalty in §3.2.1, it has to be captured at recording time.

The evaluation also needs controlled examples of the behaviours the detector is meant to find. The precision and recall results in §4.2.2 are computed against sessions where a

specific process issue was deliberately introduced. A free-form session dataset would not provide that controlled injection signal directly. For these reasons, we recorded our own dataset using the IntelliJ plugin described in §3.2.4, so that IDE events were captured as the sessions happened.

3.7.1 The 45-session injection set

Each session follows a scripted playbook scenario designed to trigger one specific kind of process issue. Examples include choosing a less favourable order for a set of refactoring steps, undoing and redoing the same chunk of code, skipping the expected test cadence, or making a manual edit where an IDE refactoring would have applied. The playbook covers all four divergence kinds: 5 sessions target ordering, 9 target rework, 13 target hygiene, and the remaining sessions are split between IDE-replay injections and clean controls where no issue is deliberately introduced.

3.7.2 Two label streams

Each session has two kinds of label, because the evaluation asks two slightly different questions. The first label records the divergence kind that was deliberately injected into the session, i.e. the behaviour the playbook was designed to trigger. This is used for the *injection prominence* analysis in §4.2.2. The second label is a separate hand-written set of all divergence kinds that should be detected in the session. This label can contain more than one kind, because a session designed to trigger one issue may also legitimately contain another. The per-kind precision and recall results in §4.2.2 use this second label set, and its inter-rater agreement is reported in §4.2.1.

The multi-label schema uses the four divergence kinds $\mathcal{K} = \{IDE\text{-}Replay, Rework, Hygiene, Ordering\}$ and each session is allowed to have more than one expected kind. For example, a session may contain both an ordering divergence and a hygiene flag.

3.7.3 Rater protocol

To check the schema independently, two annotators, denoted P1 and P2 in the results chapter, labelled all 45 sessions using the same schema. They completed this labelling before either of them took part in a user-study session. Each rater received the schema definition and the raw artefacts for every recording: the event log `events.jsonl` and the session metadata `session.json`. They did not receive the playbook prompts, our `manifest-v2.csv` labels, the Phase A `analysis-report.json`, or access to the dashboard. This kept the hand labels separate from both the injected labels and the detector output.

The event log stores the full file contents after each event rather than a compact diff. To make the sessions easier to inspect, raters were also given a small read-only helper script, `tool/scripts/session-event-diffs.py`, which prints a unified diff for each *edit_burst* and *refactoring_finished* event. The script is only helps to present the information from the events to the users in a more readable format and does not expose any extra information: it derives its output from the same `events.jsonl` file already given to the raters.

3.7.4 Cohen’s κ computation

Agreement is reported with Cohen’s κ [56], computed separately for each divergence kind in $\mathcal{K}$. For a given kind k , each session is treated as a binary judgement: either k is present in `expected_kinds`, or it is not. This gives a 2×2 table for each rater pair, from which κ is computed. The results are interpreted using the Landis and Koch thresholds [57]: $\kappa < 0.20$ is slight agreement, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, and 0.81–1.00 almost perfect. Because the dataset contains only 45 sessions, the raw agreement

counts from each table are reported alongside κ ; at this size, the yes/yes, no/no, and disagreement counts are still useful to inspect directly.

3.7.5 Treatment of disagreements

Disagreements are reported as part of the study rather than resolved by changing our labels or the raters' labels. When both raters agree with each other but disagree with our label, as with rework in session 043, the session is named explicitly in the results chapter and treated as a case where our label may be the outlier.

The rater-study results, including the per-kind κ values, raw agreement counts, and disagreement patterns, are reported in Table 4.8 (§4.2.1).

3.7.6 Divergence-detector evaluation protocol

The divergence-detector experiment runs the full analysis pipeline on each of the 45 recorded sessions using the production score weights. For each session, the detector output is compared with the session's `expected_kinds` label. A true positive means that the detector surfaced at least one divergence point of the expected kind somewhere in the session, a false negative means that an expected kind was not surfaced, and a false positive means that the detector surfaced a kind not present in `expected_kinds`. Matching is done at the session level rather than against an exact recorded step, because step boundaries depend on the plugin's edit-burst flushing and are not stable enough to treat as ground truth.

Precision and recall are reported separately for each kind. They are not collapsed into F1 or accuracy because the two error types have different practical meanings: a false positive shows the user an alternative that should not be there, while a false negative means the dashboard missed a useful opportunity. Accuracy is also uninformative because most kind-by-session cases are true negatives.

The experiment also checks whether the detector's output would be useful to a user. The first check is whether a surfaced alternative genuinely beats the user's trajectory, using the strict condition $\text{magnitude} > 0$ under the trajectory-final score definition in §3.3. The second is injection prominence: for the divergence kind deliberately introduced by the playbook, the experiment records whether the corresponding positive-magnitude divergence point is the highest-ranked point in that session.

Chapter 4

Experimentation and Evaluation

This chapter evaluates the score formula, the divergence-point detector, and the use of their feedback in complete refactoring sessions. The formula is tested through sensitivity and ablation experiments, the detector against labelled injection sessions, and the end-to-end workflow through a 30-session human study plus an agent extension on the same codebase.

§4.1 reports the score-formula experiments. §4.2 reports detector precision and recall against externally labelled divergence kinds. §4.3 reports the human and agent studies, where the outcome is whether feedback is reflected in later refactoring behaviour.

The labelled detector experiment uses the 45-session injection set and two-label scheme defined in §3.7. Because the per-kind counts are small, tables report raw counts alongside percentages. Every magnitude value in this chapter is the trajectory-final J delta defined in §3.2.1, so magnitudes are comparable across the four divergence kinds.

4.1 Testing the score formula

The score weights were chosen from a literature-supported severity ordering (§3.2.2), not fitted to an external outcome dataset. This section therefore tests whether the rankings are stable under plausible weight changes and whether the seven process-side terms contribute measurable signal, following the evaluation design in §3.6.

4.1.1 Sensitivity sweep

The first experiment measures how sensitive the divergence-point ranking is to the weights in the process-score formula. Large ranking changes under small weight perturbations would indicate that the order of divergence points depends too strongly on arbitrary parameter choices. The experiment uses the single-knob and multi-knob perturbation methods defined in §3.6.1. The main result is that the top-ranked divergence point is usually stable across all three session types, although some terms do affect the ranking in identifiable cases.

Experiment setup. For the single-knob sweep, the chosen weight is scaled by one of nine factors in $\{0.1, 0.25, 0.5, 0.75, 1.25, 1.5, 2.0, 4.0, 10.0\}$. With thirteen weights (7 process-side and 6 cleanliness-side), this gives $13 \times 9 \times |\mathcal{C}|$ perturbation runs per session set $\mathcal{C}$. The multi-knob sweep draws 200 independent samples per session using the log-normal sampling scheme defined in §3.6.1.

The experiment starts from three session sets: the 45 hand-labelled injection sessions of §4.2.2, the 30 user-study sessions of §4.3.1, and the 48 agent extension sessions of §4.3.2. For the sensitivity analysis, only sessions with at least two detected divergence points are rankable, because sessions with one divergence point have no meaningful ordering to

perturb and sessions with none have no ranking at all. This leaves 15 injection sessions, 25 user-study sessions, and 20 agent sessions in scope for the rank-based statistics reported below.

The three session sets still differ in their rankable-session profiles. The injection sessions have at most 5 detected divergence points per session and concentrate on single-kind exercises. The user-study sessions have up to 15 detected divergence points per session across more natural sessions. The agent sessions have at most 4 detected divergence points per session across the eight agent configurations, partly because the detector is designed around human IDE edit patterns rather than agent file-edit patterns (§4.3.2). The later magnitude-recovery statistics in §4.1.2 use the full session sets, since magnitudes are well-defined even for sessions with zero or one divergence point.

As defined in §3.6.1, ranking stability is reported using Kendall’s τ -b and top- N hit rate. Top-1 is defined for every rankable session, so it can be compared across all three session types. Top-3 and top-5 are reported on the user-study sessions, where the per-session rankings are large enough to make them informative. Several divergence-point kinds produce ties, so ties are broken by ascending step index in both the production and perturbed rankings.

Single-knob results. Table 4.1 reports top-1 stability across the rankable subsets of the three session types. The user-study set is the strongest test of ranking stability because it carries the largest rankable subset (25 of 30 sessions), the longest per-session rankings, and a low clamp-frozen rate. The injection rankable subset is the smallest (15 sessions); its top-1 stability is high but rests on a smaller per-session corpus. The agent rankable subset (20 sessions) sits between the two, although 20.8% of its (DP, perturbation) cases are clamp-frozen — the user and alt terminals both sit at the same $[0, 100]$ clamp value in both baseline and perturbed scorings, so the magnitude is identical by construction. On the user-study rankable subset, the top-1 recommendation is preserved in 2,823 of 2,925 perturbation cases (96.5%).

session set (rankable)	cases	$\tau = 1.0$	top-1 = 1	$\tau < 0$	clamp-frozen rate
Injection (15)	1,755	30.7%	99.0%	1.3%	8.6%
User study (25)	2,925	41.9%	96.5%	1.0%	1.8%
Agent (20)	2,340	72.4%	96.3%	3.0%	20.8%

Table 4.1: Single-knob sensitivity across the rankable subsets of the three session types (≥ 2 detected divergence points per session). The clamp-frozen rate is the share of $(DP \times \text{perturbation})$ cases where the user trajectory-final and the DP’s alt trajectory-final both sit at the same $[0, 100]$ clamp value in baseline and in perturbed; for those cases the magnitude $J(\text{alt}_T) - J(\text{user}_T)$ is identical by construction, contributing trivially to $\tau = 1$. Only 1.8% of user-study cases are clamp-frozen, so the 96.5% top-1 figure is overwhelmingly genuine perturbation response rather than clamp-driven invariance.

Sources of ranking instability. Table 4.2 reports how often each individual weight changes the top-ranked divergence point on the rankable user-study subset. The main pattern is that almost all top-1 shifts come from the process-side weights. Six of the seven process-side weights change the top-ranked item in 5.3%–10.7% of perturbation cases, with W_G having the largest single-weight effect (24 of 225 cases, 10.7%). The exception is W_{LAG} , which changes the top-ranked item only once (0.4%). This matches the leave-one-out result in §4.1.2: the lag term contributes magnitude, but does not noticeably reshape the user-study ranking. The cleanliness sub-weights have even less effect, changing the top-ranked item in at most 0.9% of cases, because they usually move the user and alternative trajectories in similar ways.

knob	top-1 changes / 225 cases	share
gain	24	10.7%
manualIde	17	7.6%
skipTests	15	6.7%
commitGap	15	6.7%
length	14	6.2%
broken	12	5.3%
lag	1	0.4%
cleanliness sub-weights: <i>coupling</i> 2 (0.9%), <i>smells</i> 1, <i>cohesion</i> 1, all others 0		

Table 4.2: Per-knob top-1 disruption counts on the user-study rankable subset (25 sessions $\times$ 9 perturbation factors = 225 cases per weight). Most top-1 shifts come from the process-side weights, especially W_G ; the lag weight changes the top-ranked item only once, and the cleanliness sub-weights have little effect.

Multi-knob Monte Carlo. The single-knob sweep shows how the ranking responds when one weight moves in isolation. The multi-knob Monte Carlo perturbs every weight together. Table 4.3 reports the resulting τ and top- N hit rates across $200 \times |\mathcal{C}|$ joint-perturbation samples for each session type.

The joint perturbation test gives a weaker stability result than the single-knob sweep. On the user-study rankable subset, the top-1 recommendation changes on 15.4% of samples, compared with 3.5% under single-knob perturbation, and mean τ falls to 0.586. The injection subset still has high top-1 stability (95.7%), but its mean τ drops to 0.385. This lower τ is partly a consequence of the shorter injection rankings: a joint perturbation can reorder many of the lower-ranked divergence points without changing the top recommendation. The agent subset sits between the two, with mean $\tau = 0.645$ and top-1 stability of 82.5%. Overall, the formula appears stable under small joint perturbations, but the ranking becomes less reliable as several weights move together.

session set (rankable)	samples	mean τ	top-1 = 1	top-3 = 1	top-5 = 1
Injection (15)	3,000	0.385	95.7%	32.9%	33.3%
User study (25)	5,000	0.586	84.6%	61.7%	59.5%
Agent (20)	4,000	0.645	82.5%	78.3%	75.0%

Table 4.3: Multi-knob Monte Carlo robustness across the rankable subsets (200 samples per session, log-normal $\sigma = \ln 2$). The user-study set carries the headline because it has the largest rankable subset and the longest per-session rankings. Multi-knob top-1 stability is ~ 12 percentage lower than the single-knob result on the same set. Top-3 and top-5 track top-1 on the agent set (78.3% and 75.0%) but drop on the user-study set (61.7% and 59.5%), where longer per-session rankings leave more lower-ranked divergence points available to reorder. The injection top-3/top-5 figures fall sharply because injection rankings have at most 5 items, so one lower-ranked swap can affect both metrics at once.

Clamp-frozen rate and weight-invariance. One possible source of artificial stability is the $[0, 100]$ clamp on the process score. A divergence point’s magnitude is $J(\text{alt}_T) - J(\text{user}_T)$, the difference between the alternative’s and the user’s trajectory-final process scores. A perturbation $W \rightarrow W'$ can leave this magnitude unchanged through clamp effects when both terminal scores are already pinned at clamp boundaries in the baseline scoring and remain pinned at the same boundaries after perturbation. The right-hand column of Table 4.1 reports the share of (DP $\times$ perturbation) cases in each rankable subset that satisfy this condition: 8.6% on injection, **1.8%** on user-study, and 20.8% on agent. The user-study figure is the smallest of the three, so almost all of the 96.5% top-1 stability on that set reflects genuine perturbation response rather than clamp-driven invariance.

The agent rate (20.8%) is materially higher, and the per-session breakdown explains why. Four of the twenty rankable agent sessions finish with $J(\text{user}_T) = 0$: the S1 sessions across the four Claude-family stacks, where the agent left the build broken at the end of S1 (the same pattern surfaced in §4.3.2, stack 1 Thread 1).

Caveats and scope. These results should be read with three limits in mind. The sensitivity experiment shows that the formula is stable near the production weights, but it does not show that those weights are uniquely correct. A stronger validation would refit the weights against an external outcome dataset, such as longitudinal code-quality measurements, but no such dataset currently exists for this problem. The multi-knob test is also centred on the production weights and only covers the neighbourhood within roughly $\pm 2\sigma$ (about $\times 0.25$ to $\times 4$). Even within that range, the formula degrades on the user-study rankable subset ($\tau = 0.586$ mean), so the robustness claim is local rather than global. Finally, the rank statistics use only the rankable subset of each set ($N = 15/25/20$). The smaller injection subset limits the power of the per-set rank comparison, although the magnitude-recovery statistics in §4.1.2 still use all 45 injection sessions. These limits are treated as threats to validity in §4.2.3 and in the discussion chapter.

4.1.2 Ablation sweep

The sensitivity sweep tests whether rankings survive small weight changes. The ablation sweep tests whether the process-side terms contribute meaningful signal at all. Following the design in §3.6.2, it evaluates all $2^7 = 128$ subsets of the seven process-side weights, giving 5,760 ablation cases per session set. The cleanliness sub-weights are left at their production values because the sensitivity sweep showed that changing them almost never reshuffles the ranking - including them would increase the number of variants from 128 to 8,192 for little additional information. The ablation reports sum-over-sum recovery on all three session sets in parallel (injection 45, user-study 30, agent 48), since magnitudes are defined regardless of how many divergence points a session contains. The cross-set rank comparison in Table 4.7 reports τ on the rankable subsets of injection (15 sessions) and user-study (25 sessions). Sum-over-sum recovery falls to exactly 0.000 on all three sets when every process term is zeroed, and rises to 1.000 under the full formula.

Monotone by active-term count. Table 4.4 groups the per-set ablation cases by the number of active process-score terms. On all three session sets, sum-over-sum recovery increases steadily from 0.000, when every process term is removed, to 1.000 under the full formula. The all-stripped row is the main sanity check: with all process terms set to zero, total magnitude is also zero on every set. This means the reported magnitudes are not coming from a baseline effect independent of the weights. The agent set recovers slightly faster than the injection and user-study sets at low active counts (for example, 0.225 at active-count = 1, compared with 0.170 and 0.190), which is consistent with more of the agent-set magnitude depending on a single term. The per-term breakdown in Table 4.5 shows this directly.

Per-term single-knob recovery. Table 4.5 isolates each process term by keeping that term active and zeroing the other six. The three session types show different patterns. In the injection set, **length** and **broken** recover the most magnitude on their own (0.328 and 0.302), followed by **manualIde** at 0.261 and **lag** at 0.132. In the user-study set, **length** is again the largest single-term contributor (0.426), but **skipTests** (0.371) is larger than **broken** (0.225), which suggests that the more natural user traces expose test-running behaviour more clearly than the controlled injection set. The agent set is different

active terms	injection sum/sum	user-study sum/sum	agent sum/sum	n (per set)
0 (all stripped)	0.000	0.000	0.000	45
1	0.170	0.190	0.225	315
2	0.327	0.353	0.414	945
3	0.473	0.498	0.570	1,575
4	0.612	0.634	0.700	1,575
5	0.745	0.765	0.810	945
6	0.874	0.889	0.909	315
7 (full)	1.000	1.000	1.000	45

Table 4.4: Sum-over-sum magnitude recovery by number of active process-score terms, reported separately for the injection (45 sessions), user-study (30 sessions), and agent (48 sessions) sets. Recovery rises monotonically on all three sets, from exactly 0.000 with every process term removed to 1.000 under the full seven-term formula. The agent set recovers slightly faster at low active counts, consistent with the single-term pattern in Table 4.5.

again: `manualIde` alone recovers the full magnitude budget (1.000), while `skipTests` and `commitGap` contribute exactly 0.000. This matches the pattern in §4.3.2: 41 of the 42 agent divergence points are *IDE-Replay*, which scores entirely through the manual-when-IDE weight W_{MI} . Finally, `gain` alone has sum-over-sum recovery of exactly 0.000 on all three sets because the user and alternative trajectories validate against the same terminal AST, so endpoint cleanliness gain is the same on both sides and cancels in the comparison.

term	injection sum/sum	user-study sum/sum	agent sum/sum
<code>length</code>	0.328	0.426	0.054
<code>broken</code>	0.302	0.225	0.405
<code>manualIde</code>	0.261	0.170	1.000
<code>lag</code>	0.132	0.014	0.117
<code>skipTests</code>	0.089	0.371	0.000
<code>commitGap</code>	0.080	0.125	0.000
<code>gain</code>	0.000	0.000	0.000

Table 4.5: Per-term single-knob recovery by session set, with each term active alone and the other six zeroed. `length` is the largest single-term contributor on both the injection and user-study sets, while `skipTests` ranks above `broken` only on user-study. The agent set is dominated by `manualIde` (1.000 alone), consistent with 41 of 42 agent divergence points being *IDE-Replay*. `gain` is 0.000 on every set because user and alternative trajectories share the same final AST.

Per-term leave-one-out recovery. Table 4.6 gives the complementary leave-one-out view: each row removes one term from the full seven-term formula. The results broadly match the single-term analysis in Table 4.5. On the injection set, removing `length` causes the largest loss of magnitude, with sum-over-sum recovery falling to 0.718. The user-study set is similar: removing `length` gives the largest drop (0.636), followed by `skipTests` (0.718). The agent set follows a different pattern. Removing `manualIde` drops recovery from 1.000 to 0.405, whereas removing `length` has little effect (0.950). Removing `skipTests` or `commitGap` has no effect on agent recovery, because those terms carry no agent-set magnitude in this experiment.

The `gain` row behaves differently from the penalty terms. Removing it increases sum-over-sum recovery to 1.060 on injection, 1.120 on user-study, and 1.108 on agent, so the absolute divergence magnitudes become larger than under the production formula. This does not mean that `gain` is a bad term. In these comparisons, the user and alternative trajectories end at the same terminal AST, so endpoint cleanliness gain is shared by both sides rather than separating them. Its main effect is therefore through the $[0, 100]$ score clamp: a shared gain value can move both scores towards the same clamp boundary and

reduce the visible gap created by the penalty terms. Removing **gain** removes some of that compression, so the remaining penalty differences can appear larger. **lag** shows a smaller version of this pattern on user-study (1.015) and agent (1.045), but not on injection (0.842).

removed term	injection sum/sum	user-study sum/sum	agent sum/sum
length	0.718	0.636	0.950
manualIde	0.822	0.876	0.405
broken	0.833	0.960	0.851
lag	0.842	1.015	1.045
skipTests	0.899	0.718	1.000
commitGap	0.943	0.900	1.000
gain	1.060	1.120	1.108

Table 4.6: Per-term leave-one-out recovery by session set. Each row keeps six terms active and removes one. **length** has the largest effect on injection and user-study, while **manualIde** is most important on agent. Removing **gain** increases the recovered magnitude above 1.000 because it removes part of the score-clamp compression described in the text.

Cross-set rerun on the user-study sessions. Table 4.7 repeats the leave-one-out test as a rank-stability comparison on the rankable subsets of the injection and user-study sessions (15 and 25 sessions respectively - see §4.1.1). On the user-study subset, **length** is the clearest rank-carrier: removing it gives the lowest τ in the column (0.527). **manualIde** and **broken** follow, while **skipTests**, **commitGap**, and **gain** have more moderate effects. Removing **lag** has the smallest user-study impact ($\tau = 0.784$), which suggests that on these longer human traces it affects magnitude more than the ordering of divergence points. The injection values are lower overall, reflecting the smaller rankable subset, but they still identify **length** and **manualIde** as the strongest rank-carriers.

removed term	injection τ ($n = 15$)	user-study τ ($n = 25$)
length	0.500	0.527
manualIde	0.500	0.588
broken	0.192	0.627
commitGap	0.374	0.655
skipTests	0.444	0.663
gain	0.478	0.683
lag	0.409	0.784

Table 4.7: Leave-one-out ranking stability on the injection and user-study rankable subsets, sorted by user-study impact. Removing **length** causes the largest user-study drop, while removing **lag** has the smallest effect.

Caveats. Several limits are worth noting. This ablation covers only the seven process-side weights in §3.2.1; the cleanliness sub-weights are excluded because the sensitivity sweep found that they had almost no effect on top-1 stability (§4.1.1). The magnitude tables use the full session sets (45 injection, 30 user-study, 48 agent), since magnitudes are defined for every session. By contrast, the rank-stability table uses only the rankable subsets (15 injection, 25 user-study), as defined in §4.1.1. Finally, the agent set’s single-term pattern is driven by the population and detector limits discussed in §4.3.2: W_{MI} recovers all agent magnitude by itself, while **skipTests** and **commitGap** contribute none. This should not be read as a general property of the score formula.

The **gain**-alone row in Table 4.5 is a special case. It stays at the all-stripped baseline (sum/sum 0.000) because the user and alternative trajectories end at the same final AST,

so endpoint cleanliness gain is shared by both sides. In the leave-one-out table, **gain** matters only through its interaction with the $[0, 100]$ score clamp.

4.2 Testing the divergence-point detector

This section evaluates the divergence-point detector against external ground truth. Because the detector is scored against per-kind labels, the first step is to check that those labels are reliable. §4.2.1 reports inter-rater agreement for the protocol described in §3.7. §4.2.2 then reports per-kind precision and recall on the 45 injection sessions, and §4.2.3 discusses a known recall weakness that is left open.

4.2.1 Inter-rater reliability

Table 4.8 reports per-kind Cohen’s κ for the three rater pairs on the 45-session set, using the labelling protocol and interpretation thresholds from §3.7. All four divergence kinds reach at least substantial agreement. Ordering and IDE-replay are unanimous across all three labellers ($\kappa = 1.00$ for every pair). Rework is also high, with $\kappa = 0.86$ between the author and each independent rater, and $\kappa = 1.00$ between P1 and P2. Hygiene is the lowest but still substantial, ranging from $\kappa = 0.72$ to $\kappa = 0.86$.

kind	Author vs P1	Author vs P2	P1 vs P2
<i>Ordering</i>	1.000	1.000	1.000
<i>IDE-Replay</i>	1.000	1.000	1.000
<i>Rework</i>	0.861	0.861	1.000
<i>Hygiene</i>	0.848	0.723	0.862

Table 4.8: Per-kind Cohen’s κ on the 45-session set for all three rater pairs. Ordering and IDE-replay are unanimous; rework and hygiene both reach at least substantial agreement.

The remaining disagreements are concentrated in a small number of sessions. For Rework, only two of the 45 sessions differ: both independent raters label session 024 (borderline suboptimal ordering) as Rework while the author does not, and the author labels session 043 (borderline add-then-revert) as Rework while neither rater does. Since P1 and P2 agree with each other in both cases, the author’s labels on these sessions may be the outliers. Hygiene disagreements are also localised. P2 labels three manual-move-method sessions (014, 015, 016) as Hygiene while the author and P1 do not, and the author labels session 039 (borderline no-commit stretch) as Hygiene while neither rater does. Following the protocol in §3.7, these disagreements are reported as findings rather than resolved by editing the labels.

4.2.2 Divergence experiment

The third experiment evaluates the divergence-point detector against ground truth, using the protocol defined in §3.7.6. Precision is perfect for all four kinds: Rework, Hygiene, Ordering, and IDE-Replay all score 1.00. Across the 45 injections, the detector surfaces an alternative in 36 cases. Of those, 26 have positive magnitude, meaning the alternative strictly beats the user’s trajectory under the production formula. Every positive-magnitude catch is ranked top-1 within its session.

Precision and recall. Table 4.9 reports per-kind precision and recall. Rework and Hygiene are perfect on both axes: all 9 Rework injections and all 13 Hygiene injections are caught with 1.00 precision. IDE-Replay is also precision-perfect. Ordering is different: precision remains 1.00, but recall is 0.36 on any-expected and 0.40 on injection-only. This

is because the reorder synthesiser’s `splitOnInvalid` validator check (§3.3.2) rejects any window containing a step the synthesiser cannot reproduce, and most Ordering-injection windows in this set contain at least one such step. This recall gap is documented and its rationale discussed in §4.2.3. The reliability of the ground-truth labels is reported in §4.2.1.

kind	precision	recall (injection)	recall (any expected)
<i>Ordering</i>	1.00	0.40	0.36
<i>IDE-Replay</i>	1.00	0.76	0.76
<i>Rework</i>	1.00	1.00	1.00
<i>Hygiene</i>	1.00	1.00	1.00

Table 4.9: Per-kind precision and recall on the 45-session set. The “injection” recall column counts only the kind the playbook explicitly injected; “any expected” counts any kind the session’s `expected_kinds` label includes.

Detection quality. Precision and recall show whether the detector finds the expected kinds, but they do not show whether the proposed alternatives are actually better than the user’s trajectory. Table 4.10 therefore looks at the quality of the emitted divergence points. For each kind, it reports how many divergence points are emitted and whether the best alternative beats, ties, or loses to the user’s trajectory under the production score formula. Across all kinds, best-alternative magnitudes range from -2 to $+23$, with a mean of $+6.07$.

Hygiene is the strongest case: all 13 Hygiene divergence points strictly improve on the user’s trajectory. IDE-Replay also performs well, with 12 improvements from 16 divergence points (75.0%). Of the four non-improvements, three are tied at zero because the user’s score is already pinned at the bottom of the $[0, 100]$ range, so the IDE-driven alternative cannot score lower either. The remaining case is a genuine loss, where the IDE alternative scores 5 points below the user’s manual edit. These misses mostly come from the score floor: in three cases the user’s score was already clipped at the bottom of the $[0, 100]$ range, so the IDE alternative could not show an improvement even if it avoided the manual-IDE penalty.

Rework is more mixed. It beats the user in 6 of 13 divergence points (46.2%), ties once, and loses six times. This is partly because the rework synthesiser is deliberately simple: it removes add-then-undo code chunks without modelling why the intermediate states existed. In practice, a user may add a guard, run tests, decide it is unnecessary, and remove it. The final diff looks like wasted motion, but the temporary code may have kept later intermediate states compiling or passing tests until the rest of the refactor caught up. Removing it just because it disappears later can therefore introduce extra broken-build checkpoints in the alternative path. This is also a methodological point: undoing work is not always bad. A slightly longer path that verifies intermediate correctness can score higher than the shortest path to the same final state. The Rework beat rate therefore separates cases where the removed code chunk really was wasted motion from cases where it helped preserve a better intermediate trajectory, while also showing the limit of a simple chunk-stripping approach.

Ordering emits 24 divergence points after considering 194 valid permutations. Each divergence point corresponds to one reorder window, and its magnitude is taken from the best permutation found for that window. Ten windows (41.7%) beat the user’s order, 12 tie at zero, and 2 are worse than the user’s order. The important point is that these alternatives must have an identical final AST as the user’s path, so endpoint terms alone cannot separate them. The production formula separates them through W_{LAG} , which rewards paths that improve intermediate cleanliness earlier. In the lag ablation, only 4 Ordering windows

remain positive, and the two negative cases collapse to zero.

kind	DPs	beats user	ties user	loses to user	beats fraction	max magnitude
<i>Rework</i>	13	6	1	6	46.2%	8 points
<i>Hygiene</i>	13	13	0	0	100%	9 points
<i>IDE-Replay</i>	16	12	3	1	75.0%	23 points
<i>Ordering</i>	24	10	12	2	41.7%	9 points

Table 4.10: Per-kind detection quality. “DPs” gives the number of emitted divergence points. The *beats*, *ties*, and *loses* columns group those points by the sign of $J(\text{alt}_T) - J(\text{user}_T)$ under the production formula. *Hygiene* improves on the user in every case; *Ordering* has many ties because several reordered alternatives reach the same final state with the same final score.

Ordering can also favour the user. Two of the 24 Ordering windows have negative magnitude under the production formula, meaning the user’s original order scores better than the synthesised alternative. Session 043 is the clearest example: both paths reach the same final state, but the alternative has worse intermediate cleanliness, so it accumulates more lag penalty. This shows that W_{LAG} is not just increasing positive Ordering magnitudes; it also allows the formula to recognise cases where the user’s order was the better one.

Injection prominence. Table 4.11 reports how prominently the injected problem appears in the detector output. The detector catches 26 injections with a positive-magnitude alternative: 8 Hygiene, 12 IDE-Replay, 5 Rework, and 1 Ordering. In every one of these cases, the injected behaviour is the highest-magnitude divergence point in its session, giving a mean rank of 1.00 across all four kinds. A user who opens only the top dashboard recommendation in §3.5 would therefore see the deliberately injected problem in every positive-magnitude catch.

This result is partly a consequence of the controlled-injection design. Each playbook session contains one deliberately introduced problem, so that problem is likely to dominate the session’s ranking. In uncontrolled real-world sessions, several issues may be present at once, and the top-ranked item may not correspond to one clear expected kind. We therefore do not report a prominence number for the participant sessions in §4.3.1. Instead, we report their per-session, per-kind distribution.

injected kind	n caught	@ rank 1	mean rank
<i>Hygiene</i>	8	8	1.00
<i>IDE-Replay</i>	12	12	1.00
<i>Ordering</i>	1	1	1.00
<i>Rework</i>	5	5	1.00

Table 4.11: Injection prominence: rank of the caught injection’s divergence point within its session’s magnitude ordering. Every positive-magnitude catch is ranked top-1 across all four kinds.

Caveats. A caveat worth noting when interpreting these divergence numbers is that the dataset contains 45 hand-recorded sessions, so some categories have few examples: for example, there are only 5 ordering injections. The ratios are therefore best read as directional evidence, not as estimates with tight confidence intervals, and raw counts are reported alongside percentages for this reason.

4.2.3 Known limitation: ordering recall

Ordering has the weakest recall of the four divergence kinds: 0.36 on any-expected and 0.40 on injection-only. This mainly comes from the `splitOnInvalid` validator check in §3.3.2. If a reorder window contains a step that the synthesiser cannot reproduce, the window is split into single-step windows and no permutations are generated.

This matters because ordering often changes process score under the production formula. Commuting refactorings may reach the same end state, but their intermediate cleanliness trajectories can still differ. The lag term W_{LAG} assigns different magnitudes to those trajectories: Table 4.10 shows that 10 of 24 admitted ordering windows have positive magnitude under the production formula, compared with only 4 when W_{LAG} is removed. In other words, windows currently excluded by `splitOnInvalid` cannot be assumed to have zero magnitude; some would likely become positive findings if the synthesiser supported more step kinds. Closing this gap therefore depends on extending the synthesiser engine, as discussed as future work in Chapter 6. Until then, the ordering recall should be read as a lower bound on what the detector could surface with a more complete engine, rather than as a limitation of the score formula itself.

4.3 User and agent studies

§4.1 and §4.2 test the score formula and the detector in isolation; this section tests the tool end-to-end. The end-to-end question is whether the feedback the dashboard surfaces - divergence points ranked by score-delta magnitude, with concrete alternative trajectories attached - actually changes behaviour for users who did not build the tool. §4.3.1 reports a 30-session user study with five human participants on a new codebase, split into a with-feedback arm (three participants) and a no-feedback baseline arm (two participants). §4.3.2 reports an 8-stack, 48-session agent extension on the same codebase, included as a feasibility check on the same tooling rather than a co-equal study; the framing of that section, and the reasons the human-trained detectors do not map cleanly onto agent traces, are stated there.

4.3.1 User study

The 30-session user study steps outside the controlled-injection sessions to three questions. First, on sessions recorded by independent participants on a different codebase and without the playbook’s controlled-injection structure, does the detector’s per-kind output remain coherent? Second, do per-session divergence-point rates evolve as participants accumulate tool feedback across a multi-session arc? Third, is the trajectory consistent with the dashboard feedback driving it, or would the same trajectory have appeared without that feedback? The third question is what the baseline arm exists to answer.

Five participants, referred to throughout as P1 through P5, were recruited as senior MEng Computing students at Imperial College London. All five performed six recorded refactoring sessions each on the same order-processing codebase (`user-study-fixture/`), structurally disjoint from the library codebase used in §4.2.2: 30 sessions in total. P1, P2, and P3 are the with-feedback arm: at the end of each session they viewed the dashboard’s divergence points and trajectory advice for the session they had just completed before starting the next one. P4 and P5 are the no-feedback baseline arm: they performed the same six-session playbook on the same codebase, but never saw the dashboard between sessions. Every other aspect of the protocol - the playbook prompts, the codebase, the plugin instrumentation, the reset script between sessions - was identical across arms. Two of the five participants (P1 and P2) also produced an independent labelling of the 45-session injection set before performing their own user-study sessions; that labelling, together with

our own, feeds the three-rater inter-rater agreement reported in §4.2.1 and is not repeated here.

Arm assignment was randomised: two of the five participants were drawn uniformly at random to form the baseline arm before any participant began their sessions. The arm sizes ($n = 3$ vs $n = 2$) are too small to support hypothesis testing, and no p -value is claimed in this section. What the design does admit is a direction-of-effect comparison: whether the gain-stripped process score and the per-session divergence-point counts move in the same direction in both arms, or diverge.

Descriptive findings on the participants’ sessions. The playbook gives each session a what-not-how prompt naming the area of code to address but never the IntelliJ action to use, the testing or commit cadence to follow, or the divergence kinds the detector surfaces.¹ Because the playbook deliberately omits pre-declared `expected_kinds`, the participants’ sessions admit no precision/recall claim of the form reported in §4.2.2; what they admit is a descriptive per-kind distribution and a descriptive per-session count, which Table 4.12 gives in aggregate by arm and by participant.

arm	participant	<i>IDE-Replay</i>	<i>Rework</i>	<i>Hygiene</i>	<i>Ordering</i>	total	DP / ses
with-feedback	P1	12	2	5	2	21	
with-feedback	P2	9	0	8	1	18	
with-feedback	P3	5	0	5	0	10	
with-feedback (sub)	$n = 18$ sessions	26	2	18	3	49	
baseline	P4	21	3	16	0	40	
baseline	P5	13	0	17	2	32	
baseline (sub)	$n = 12$ sessions	34	3	33	2	72	
combined	$n = 30$ sessions	60	5	51	5	121	

Table 4.12: Per-participant divergence-point counts across the 30-session user study, grouped by arm. All five participants registered 0 divergence points on session S04 (a cross-class duplication consolidation task that all five executed as straight manual edits) and on the small fraction of other sessions where the playbook prompt closed without a reorder window, an IDE-replayable manual sequence, or a hygiene flag. Full per-session breakdowns are in the reproducibility notebook (`tool/fixtures/notebooks/experiments.ipynb`); the per-participant totals above are the cleanest summary of the per-kind distribution.

The aggregate per-kind distribution across the 121 divergence points is IDE-replay 49.6%, hygiene 42.1%, rework 4.1%, ordering 4.1% - broadly consistent with the kind proportions seen on the 45-session injection set in §4.2.2, where IDE-replay and hygiene also dominated. Both arms exercise all four kinds, with rework and ordering sparse in both. The cleanest single number in the table is the per-session divergence-point rate: 2.7 DPs per session in the with-feedback arm against 6.0 in the baseline arm, a $2.2\times$ difference. The arm split also persists session-by-session - the trajectory table below shows that both arms start at comparable totals at S1 and diverge across the arc, rather than the baseline arm running at a higher rate uniformly. All five participants registered 0 divergence points on session S04, the cross-class validation consolidation prompt, which asks the participant to consolidate three near-identical validation blocks into a single home (a delete-an-inline-block plus delete-a-dead-method task). All five executed it as straight manual edits with no rework, no skipped tests, no manual-where-IDE-could-replay candidate, and no reorder window, so the detector legitimately had nothing to surface. Used cautiously, this is the closest the chapter comes to an external true-negative observation: when the task can be discharged in a couple of clean edits and the participants do so, the detector stays quiet

¹Quotes throughout this section are paraphrased from session-end notes taken during the post-study interview; full audio transcription was not feasible within the study budget. The source notes are retained as private study artefacts and are not reproduced here.

on all five of them. Both with-feedback participants from the original cohort identified IDE-replay as the most actionable kind in interview, with one noting that they “didn’t realise that IntelliJ could do all of these refactorings” so didn’t use them; both identified ordering as the least actionable, with one asking “how can I actually apply that in the future without knowing”.

Behavioural trajectory across the six-session arc. The trajectory question is whether per-session divergence-point counts evolve as participants accumulate feedback - and, with the baseline arm in place, whether the same evolution is also present in participants who never saw the dashboard. Table 4.13 aggregates divergence-point counts session-by-session within each arm. The with-feedback aggregate falls from 12 in S1 to 2 in S6, an 83% reduction with a mid-arc peak at S2 driven by hygiene, then a near-monotonic decline. The baseline aggregate starts at 10 in S1 - comparable to the with-feedback arm’s 12 at the same session - and ends at 15 in S6, having risen rather than fallen. The two arms diverge from S3 onwards: the with-feedback arm drops from 11 at S3 to 2 at S6, while the baseline arm rises from 19 to 15 across the same range (with an S5 trough at 13). The participants in the with-feedback arm self-reported the corresponding behavioural change in interview, with the original cohort describing how they “ran tests after every change and committed more and used the provided automated tools” and how they “used the IDE refactoring methods more frequently”. The arm-level pattern aligns with those reports.

arm	kind	S1	S2	S3	S4	S5	S6	Δ
with-feedback	<i>IDE-Replay</i>	8	6	6	0	5	1	-7
with-feedback	<i>Hygiene</i>	3	11	2	0	2	0	-3
with-feedback	<i>Rework</i>	0	0	2	0	0	0	0
with-feedback	<i>Ordering</i>	1	0	1	0	0	1	0
with-feedback (sub)	total	12	17	11	0	7	2	-10
baseline	<i>IDE-Replay</i>	5	3	11	0	7	8	+3
baseline	<i>Hygiene</i>	4	11	7	0	6	5	+1
baseline	<i>Rework</i>	0	1	1	0	0	1	+1
baseline	<i>Ordering</i>	1	0	0	0	0	1	0
baseline (sub)	total	10	15	19	0	13	15	+5

Table 4.13: Per-kind divergence-point trajectory across the six-session arc, summed within each arm. The with-feedback aggregate drops from 12 at S1 to 2 at S6 ($\Delta = -10$); the baseline aggregate rises from 10 to 15 ($\Delta = +5$). The two arms are comparable at S1 and diverge from S3 onwards. Both arms register 0 divergence points on S04 (a clean-consolidation prompt all participants discharged in a couple of edits). IDE-replay and hygiene carry most of the arm-level signal in both directions.

The two arms differ on which kinds carry the trajectory. In the with-feedback arm, IDE-replay falls from 8 to 1 and hygiene from 3 to 0 (with the S2 hygiene spike consistent with participants first seeing commit-cadence feedback at the end of S1 and applying it from S3 onwards). In the baseline arm, IDE-replay does not fall - it ends +3 on the arc - and hygiene ends +1 rather than -3. Rework and ordering are episodic in both arms: ordering fires rarely on the participants’ sessions (5 positive-magnitude divergence points in total across 30 sessions), and rework spikes only on the hardest sessions where contiguous undo/redo cycles drive several flags from a single sequence of mis-clicks. One with-feedback participant raised the rework behaviour as a metric-design complaint in interview, observing that rework “picked up on undos + redos a lot (in a bad way) [-] over penalised” on contiguous-step sequences caused by mistyped keys; the per-session detail behind the trajectory table is consistent with that account.

The six sessions are not interchangeable: S2 (magic-number renames) and S4 (cross-class consolidation) are short, mechanical sessions where divergence-point counts collapse

for any participant regardless of arm, while S3 and S6 are structurally harder sessions where the opportunity for hygiene and rework flags is built into the playbook prompt. Cross-session comparisons of raw divergence-point counts within a single arc are therefore confounded by task difficulty - which is exactly why the baseline arm matters. The baseline arm rides the same task-difficulty envelope as the with-feedback arm (both arms see the same S2 hygiene rise and the same S4 quiet), so the divergence between the two arms across the arc is the signal that is left after the task-difficulty confound is held constant. The same confound applies to the agent extension in §4.3.2.

Process scores under the production and gain-stripped weightings. The within-session process score itself can be partially separated from task difficulty by inspecting it twice: once under the production weights, and once under a weighting that sets the cleanliness-gain weight W_g to zero. The cleanliness-gain term is the only term in the score formula that scales with how much the code itself improved over the session, and the ablation in §4.1.2 already showed that this term acts as a regularizer on divergence-point ranking rather than reshaping it. W_g is the largest weight in the production formula by design, since the anti-gaming axiom of §3.2.1 requires the cleanliness reward to outweigh any process-side shortcut a developer could plausibly take by trading off code quality for procedural discipline; zeroing W_g in a free-form deployment would invite exactly that gaming. The participants’ sessions are not free-form: the playbook fixes the structural change each session must produce, so the cleanliness outcome is set by whether the participant completed the prompt rather than by anything the participant could trade off, and the gain-stripped view is a legitimate read on the process-discipline terms alone for this experiment. Zeroing W_g leaves the broken-time, skipped-tests, manual-when-IDE, length-bonus, and commit-gap terms in place, so the remaining score asks “did the user execute the process well” rather than “did the code end up cleaner”. Table 4.14 reports the gain-stripped score across the six-session arc for all five participants, grouped by arm.

arm	participant	S1	S2	S3	S4	S5	S6	Δ	slope
with-feedback	P1	25	22	41	38	34	48	+23	+4.6
with-feedback	P2	16	13	29	39	37	47	+31	+6.2
with-feedback	P3	39	15	42	42	31	48	+9	+1.8
with-feedback (mean)	($n = 3$)	26.7	16.7	37.3	39.7	34.0	47.7	+21.0	+4.2
baseline	P4	22	17	8	22	12	15	-7	-1.4
baseline	P5	23	17	9	36	11	20	-3	-0.6
baseline (mean)	($n = 2$)	22.5	17.0	8.5	29.0	11.5	17.5	-5.0	-1.0

Table 4.14: Trajectory-final process score across the six-session arc for each participant under the cleanliness-gain-zeroed weighting ($W_g = 0$), grouped by arm. The production-weighted view is dominated by the cleanliness-gain term and is reported per-session in the reproducibility notebook. Δ is $S6 - S1$ and slope is $\Delta/5$ in score units per session. The arm means aggregate across participants within the arm.

Two patterns sit in the table. First, all three with-feedback participants finish above where they started (+23, +31, +9), while both baseline participants finish below where they started (-7, -3). The arm means separate cleanly: +21.0 vs -5.0, or +4.2 per session vs -1.0 per session. With per-participant variance comparable to the between-arm difference, the finding is reported as a direction of effect rather than a hypothesis test - which is what $n = 3$ vs $n = 2$ admits.

Second, the within-arc shape carries the same task-difficulty signature in both arms: S2 dips in four of five participants, S4 sits near a per-participant ceiling for everyone, and the cross-arm separation builds from S3 onwards. S2 (magic-number renames executed under intermediate-cleanliness pressure) and S3 (within-class duplication consolidation under

stricter test-and-commit cadence) are the two sessions where the dashboard’s hygiene and IDE-replay advice would have been most actionable for a participant who saw it after S1, which is consistent with the with-feedback arm’s S3 mean (37.3) sitting above the baseline arm’s (8.5) by the largest single-session margin on the arc. The S4 convergence (39.7 vs 29.0) reflects that S4 is a clean-execution session for everybody and offers little room for either arm to differentiate.

The playbook orders the six sessions in roughly increasing structural complexity (S1 renames through to S6 multi-step reshape). The with-feedback arm’s improvement is therefore happening despite progressively harder tasks, not riding on easier ones - and the baseline arm’s decline is happening over the same difficulty curve, which removes increasing difficulty as a confound for the arm-level direction. P3’s lower slope (+1.8) is consistent with starting at a higher S1 (39) closer to ceiling than P1 or P2: P3 has less room to improve, not a different arc shape.

With-feedback vs no-feedback baseline arm. The arm-level comparison is the strongest causal-direction signal in the chapter, because arm assignment was randomised over a single participant pool and every other aspect of the protocol was held constant. The baseline arm rides the same task difficulty, the same codebase, the same playbook order, and the same plugin instrumentation as the with-feedback arm; the only manipulated variable between arms is whether the participant saw the dashboard between sessions. The arm means in Table 4.14 report a +21.0 vs −5.0 gain-stripped Δ across the arc (a 26.0-point gap, or +4.2 vs −1.0 per session), and the arm-level totals in Table 4.13 report a −10 vs +5 divergence-point Δ across the arc (a 15-DP gap). Both signals point the same direction.

The gain-stripped subscore is largely insulated from how familiar the participant is with the codebase itself - the terms active under $W_g = 0$ are commit cadence, skipped tests, and manual-when-IDE rate, none of which depend on knowing the code better. What the design does not separate is the with-feedback participants intuiting what the dashboard rewards and adjusting behaviour to satisfy it (a score-gaming reading), or becoming more comfortable with the playbook idiom across the arc (a task-style-familiarity reading). The threats-to-validity chapter (§5.5) discusses both. The baseline arm rules out one strong alternative reading - that the with-feedback arc reflects the playbook getting easier as the participant moves through it - because the baseline participants worked the same playbook in the same order and trended downward across the arc rather than upward. The remaining limit is sample size. With $n = 3$ vs $n = 2$ the arm means are estimated from too few participants to support a hypothesis test, and per-participant variance is comparable to the between-arm difference, so the result is reported as a randomised but underpowered direction-of-effect claim: “under random arm assignment, the with-feedback arm improved across the arc, the baseline arm did not, and the within-arm variance is smaller than the between-arm difference.” No p -value or confidence interval is claimed.

One additional qualitative finding does not show in the trajectory tables but matters for future tuning: hygiene also produces a false-positive shape, firing when a participant is thinking rather than stuck. One with-feedback participant flagged this in the interview as being “penalised for not running tests when wasn’t quite finished but just was thinking”, and the pattern is reflected in the S2 hygiene peak in both arms. Both this finding and the rework over-penalisation are reported here as candidate metric-tuning directions rather than detector bugs: the divergence points are accurate readings of what the trajectory contains, but the metric design treats some legitimate-pause and mistyped-key cases more harshly than the participant would.

Per-step inspection as a usability win. Both participants singled out the dashboard’s per-checkpoint detail panel as a positive part of the interface, independently of the four

detector kinds. One participant said the per-checkpoint view made it easy to “see what action you made to cause the issues to happen”, and that the trajectory graph showing the highest score reachable from each point onwards was “egging you on to be a better person” (i.e. the alternative-trajectory overlays were read as constructive rather than punitive). The other participant said they could “see the specific point you went wrong”, valued the cleanliness-signal “breakdowns” on each step, and called the dashboard “easy to track session at specific points and code smells”. These observations validate the per-step smell-attribution and metric-shift breakdown described in §3.2.6 as a usability feature the participants engaged with, not just an architectural by-product of how the report is assembled.

4.3.2 Agent extension experiment

Framing. This section reports an extension experiment, not a co-equal evaluation. The tool described in this thesis is designed against human edit patterns: it captures IDE refactor-menu invocations, debounces edit bursts on a 2-second idle, and thresholds commit cadence against the 60-second composite window of [5]. None of these signals map cleanly onto how a coding agent emits file changes (§4.3.2 below itemises the three structural biases that follow). The intent of running the playbook against eight agent stacks is therefore narrower than for the user study: to observe how the human-trained detectors behave on a structurally different actor, and to report what the corpus admits at face value without overclaiming the tool as an agent-evaluation framework.

Setup. Eight agent stacks ran the same six-session playbook used by the human participants, on the same `user-study-fixture/` codebase: 48 sessions in total. The eight stacks combine three model families (Claude, OpenAI GPT, Gemini) with three command-line harnesses (Claude Code, OpenCode, Cursor CLI), and include two no-feedback baselines, matching the human study’s arm structure. Six stacks are the with-feedback arm: between sessions, the human facilitator pasted the markdown output of the `feedbackForAgent` task (`tool/analysis/.../experiment/FeedbackForAgent.kt`) into the agent’s prompt at the start of the next session. That task renders an analysis report’s divergence points and trajectory advice into the same plain-text summary that a human participant would read on the dashboard, so the agent received the same information the human participants did, just in textual rather than rendered form. Two stacks are the no-feedback baseline arm (`claude-4.7-opus-medium-claude-code-baseline` and `gpt-5.5-medium-opencode-baseline`): they ran the same six-session playbook on the same codebase but were never shown the dashboard between sessions. Every other aspect of the protocol matched the with-feedback agent arm. Each agent honoured a small setup contract: tests were invoked via a wrapper script that brackets the gradle call with `TEST_RUN_STARTED` / `TEST_RUN_FINISHED` events in the plugin trace, and commits used standard `git commit` which the plugin’s reflog watcher captured as `GIT_COMMIT` events. Apart from starting and stopping the recording at session boundaries and pasting the previous session’s feedback into the next prompt where applicable, no human intervened during a session.

Per-stack descriptive arc. Table 4.15 reports the gain-stripped trajectory-final process score across the six-session arc for each of the eight agent stacks, along with the per-stack divergence-point total. The gain-stripped view is used here for the same reason as the human user-study table: it isolates the process-discipline terms from the cleanliness-gain term, which is dominated by playbook completion rather than process quality.

Four observations follow. First, every agent stack sits at $J = 45$ at S4 - a uniform-across-stacks corroboration of the user-study finding that S4 is genuinely a no-event session under the playbook. Second, all eight stacks finish above where they started: the smallest

#	stack	arm	S1	S2	S3	S4	S5	S6	ΔJ
1	claude-2-opus-4.7-claude-code	feedback	35	39	40	45	40	35	0
2	claude-opus-4.7-medium-opencode	feedback	37	36	40	45	37	38	+1
3	claude-opus-4.7-thinking-medium-cursor-cli	feedback	35	39	40	45	39	39	+4
4	gemini-3.5-flash-medium-opencode	feedback	33	37	40	45	39	37	+4
5	gpt-5.5-medium-cursor-cli	feedback	23	38	39	45	37	37	+14
6	openai-gpt-5.5-medium-opencode	feedback	35	38	40	45	36	36	+1
7	claude-4.7-opus-medium-claude-code-baseline	baseline	36	34	39	45	36	40	+4
8	gpt-5.5-medium-opencode-baseline	baseline	25	34	40	45	36	38	+13
with-feedback (mean, $n = 6$)			33.0	37.8	39.8	45.0	38.0	37.0	+4.0
baseline (mean, $n = 2$)			30.5	34.0	39.5	45.0	36.0	39.0	+8.5

Table 4.15: Per-stack gain-stripped trajectory and total divergence-point count across the eight agent stacks. Stacks 1–6 are with-feedback (named for the model $\times$ harness combination); stacks 7 and 8 are no-feedback baselines, named with the **-baseline** suffix. The arm means aggregate across stacks within the arm.

ΔJ in the table is 0 (stack 1) and the largest is +14 (stack 5). Third, the divergence-point counts are small and almost entirely IDE-replay: across 48 agent sessions the total is 42 DPs, of which 41 are IDE-replay, 1 is ordering, and 0 are rework or hygiene. Fourth, the per-stack DP rate (0.7–1.5 per session for the with-feedback arm, 0.5–0.7 for baseline) is an order of magnitude lower than the human cohort’s (2.7–6.0 per session); agents start at a much higher procedural-discipline floor than the human participants do.

With-feedback vs no-feedback baseline arm. The arm-level comparison in Table 4.15 does not show the same direction-of-effect signal that the human study did. The with-feedback arm gains +4.0 on ΔJ across six stacks; the baseline arm gains +8.5 across two stacks. Both arms’ slopes are positive, but the baseline arm’s slope (+1.7 per session) is larger than the with-feedback arm’s (+0.8). Per-session DP totals tell the same story: the with-feedback arm sums to 36 DPs across 36 sessions (1.0 per session), the baseline arm to 7 DPs across 12 sessions (0.6 per session). On this corpus the tool’s between-session feedback shows no measurable causal effect on agent process scores, and what difference does appear runs in the opposite direction to the human result.

Several factors plausibly contribute to the null result, none of them indicating the tool’s feedback is harmful to agents:

- **Floor effect.** Agents start at a much higher procedural-discipline ceiling than humans (mean S1 gain-stripped $J \approx 33$ across stacks vs ≈ 22.5 for the human baseline arm, with baseline DP rates ≈ 0.5 /session for agents vs ≈ 6.0 /session for human baselines). There is materially less room for any intervention to move the score.
- **Detector-bias floor.** Three of the four divergence-point kinds rarely fire on agent traces at all (see §4.3.2 below). Whatever between-session adjustment a with-feedback agent makes on those kinds is invisible to the score by construction.
- **Anomalous S1 starts in the baseline arm.** The $n = 2$ baseline arm includes **gpt-5.5-medium-opencode-baseline** starting at $S1 = 25$ (the lowest S1 in the table), which inflates that stack’s ΔJ to +13. With one of the two baseline stacks starting that far below ceiling, the baseline arm mean is sensitive to a single low-S1 reading.

Cross-stack effects. The six with-feedback stacks span a model axis (Claude vs Gemini vs GPT) at fixed harness and a harness axis (Claude Code vs OpenCode vs Cursor CLI) at fixed model. With one stack per axis position there is no inferential weight to give

the comparison, but the descriptive shape is worth recording. On the model axis with the OpenCode harness held out, Claude (stack 2), Gemini (stack 4), and GPT (stack 6) produce slopes $+0.20/ +0.80/ +0.20$ and DP totals 4/9/5: Gemini surfaces the most DPs but climbs J the fastest, consistent with a willingness to commit mid-task partial work that surfaces as IDE-replay but recovers by S6. On the harness axis with the Claude model held out, Claude Code (stack 1), OpenCode (stack 2), and Cursor CLI (stack 3) produce slopes $0.00/ +0.20/ +0.80$ and DP totals 5/4/6: Cursor CLI improves fastest at fixed model. The cross-stack comparison (stack 1 vs stack 6, Claude / Claude Code at slope 0.00 vs GPT / OpenCode at slope $+0.20$, 5 DPs each) is roughly tied.

Detector-bias limitations on agent traces. Three structural reasons explain why the human-trained detectors under-fire on agent traces:

- **No IDE-driven refactorings.** Agents apply file edits via text tool calls, not via IntelliJ’s Refactor menu. Every structural refactoring an agent performs is classified as a manual edit by the detector, so the *IDE-Replay* kind fires on the structural refactorings that an IntelliJ-side refactor would have produced - but the agent has no actuator to switch to even if it reads the feedback. This is the dominant kind in the agent corpus (41 of 42 DPs) and is the cleanest example in the chapter of feedback being correctly interpreted but structurally inactionable.
- **Commit-gap threshold rarely triggered.** The *Hygiene* kind’s commit-gap detector flags six refactor checkpoints elapsed without a commit. Agents batch many file changes per checkpoint and arrive at S6 in fewer total checkpoints than a human session of equivalent length. In a six-session arc the threshold is rarely reached, which is consistent with the 0 hygiene DPs across all 48 agent sessions.
- **Edit-burst debouncer collapses each agent turn into one burst.** The edit-burst tracker debounces on `DEBOUNCE_MS = 2000L` (2 s; see `tool/ide-plugin/.../EditBurstTracker`). It does fire on tool-applied file writes. But agents emit many file edits in a tight burst, then pause for reasoning longer than 2 s; each tool turn therefore collapses to one burst event, compressing the burst-vs-checkpoint signal that the *Rework* and *Ordering* kinds depend on. This is consistent with the 0 rework DPs and 1 ordering DP across the 48 agent sessions.

None of these are tool bugs - the detectors do the right thing on human traces. They imply that the agent results in Table 4.15 should be read as descriptive feasibility evidence rather than a verdict on the tool’s intended use, which is human-developer refactoring.

Feedback engagement varies by model family. The aggregate arm comparison says feedback did not move the score. Per-stack transcripts say something more interesting. Skimming the `transcript.md` file at the root of each with-feedback stack directory shows a sharp split. The three Claude-family stacks (1, 2, 3) explicitly engaged with the feedback in writing. Stack 1 is the strongest case: after S1 warnings about `BUILD_OFTEN_BROKEN`, `TEST_REGRESSIONS`, and `PROCESS_SCORE_DEGRADED`, S2 opens with “commit smaller and more often - one commit per rename (or per logical step), not one at the end. Test after each step.” After S3 flagged `REFACTORING_INTRODUCED_SMELLS`, S4 planning included “when extracting a method, leave the extracted body cleaner - extract and tidy, not extract and stop.” Stack 2 (`claude-opus-4.7-medium-opencode`) follows the same pattern: after S1, “`BUILD_OFTEN_BROKEN` - likely caused by intermediate states - I’ll land each rename atomically per checkpoint.” Stack 3 (`claude-opus-4.7-thinking-medium-cursor-cli`) shows quieter engagement but visible adjustment after the S5 IDE-replay summary. The

three non-Claude stacks (4 Gemini, 5 GPT-5.5 Cursor CLI, 6 GPT-5.5 OpenCode) show no visible engagement: they proceed with consistent (and competent) behaviour session-to-session but do not acknowledge or react to the between-session feedback in their conversation traces.

The split is descriptively interesting on its own, and is a reasonable starting point for a follow-up study on how different model families treat reflective critique in iterative tasks. It does not, however, rescue a causal claim about the tool’s feedback driving score improvement. Even on the Claude-family stacks, where the transcripts show the feedback being read and acted on, the gain-stripped ΔJ does not separate from the baseline arm.

Qualitative threads from stack 1 (claude-2-opus-4.7-claude-code). The aggregate trajectory table flattens what is, on inspection, a substantively richer story for stack 1 specifically. The agent’s written reflections after each session show it reading the feedback, diagnosing its own anti-patterns, and adjusting its approach between sessions in ways the per-kind counts only partially capture. These threads are reported as stack-1 anecdotes that illustrate what feedback engagement looks like in transcript form, rather than as findings that generalise across the eight-stack agent extension.

Thread 1: commit cadence, over-corrected and then resolved. S1 produced stack 1’s worst-scoring session of the arc: one bulk commit at the end of all three renames, a session-long stretch without committing, a 28% build-broken rate (the build was broken on 2 of 7 checkpoints), one test regression, and a production-weighted process score that fell from 50 to 0 - the largest single-session drop on the arc. The post-session feedback fired three advice items at critical and warning levels. The agent’s response, written before S2 began, was the direct procedural correction quoted above. S2 implemented exactly this: three steps as three separate `git commit` calls, each preceded by `./agent-test`; the resulting session produced 0 divergence points, 0 advice items, and the production-weighted process score climbed from 50 to 79. The agent then over-corrected in the opposite direction. Its pre-S3 statement was “I’ll aim to commit even more granularly (each individual constant/edit, not each step)”, and S3 fired three IDE-replay divergence points (one at magnitude -6), `REFACTORING_INTRODUCED_SMELLS`, and a production-weighted score drop from 50 to 30. By S6 the agent had revised the rule a second time, but in a more interesting direction: the S6 advice flagged an 8-checkpoint commit stretch and a 37% build-broken rate (3 of 8 checkpoints), and the agent’s diagnosis identified the cause as a structural tension the rule itself created: “for an Extract Method, do it as a single atomic edit (insert new method first, then replace call site) so the build never breaks mid-step.” The agent had discovered, by being penalised for it twice, that “commit more often” and “don’t break the build” are in conflict when an edit is structurally multi-step, and resolved the conflict by switching to larger atomic diffs rather than finer-grained commits.

Thread 2: smell-introduction feedback produced a specific self-critique. After S3 the agent fired `REFACTORING_INTRODUCED_SMELLS` alongside the process-score drop, and the agent’s written response identified its own anti-pattern by name: “methods like `chargeCash/refundVoucher` are arguably trivial wrappers that didn’t earn their keep,” and “could have made `formatAddress` cleaner.” The follow-on sessions support the diagnosis: S4 produced zero divergence points and zero advice items, S5 produced two IDE-replay flags but no `REFACTORING_INTRODUCED_SMELLS`, and the smell-introduction advice did not fire again until S6 - where the agent attributed it to multi-edit extract-method mechanics rather than the trivial-wrapper anti-pattern from S3.

Thread 3: the agent acknowledges the structural IDE-replay limitation. The persistent failure mode for stack 1 is IDE-replay. All five of stack 1’s divergence points are IDE-replay flags, and the count does not decline across the arc; it concentrates on the structurally heavier sessions (S3, S5, S6) where the agent performed enough structural refactoring to

surface any divergence points at all. The agent itself acknowledges the structural nature of this in its S6 reflection, observing that “the IDE-replay flags persist - manual extracts consistently underperform the IntelliJ refactoring; in a real run I’d use the Extract Method action.” The signal is real and the recommendation correct, but the agent lacks the tool to execute it - which is the same general observation that §4.3.2 states as a population-level structural limitation rather than a single-stack anecdote. A setup that exposed IntelliJ’s refactor actions as agent tool calls would close the IDE-replay gap; that work is named in the future-work section as the main follow-up for the agent extension.

Chapter 5

Threats to validity

This chapter collects the limitations mentioned throughout Chapter 4 and discusses design choices that affect the validity of our claims. The discussion follows the standard four-axis framework from empirical software engineering [58]: construct validity, internal validity, external validity, and statistical conclusion validity. A per-experiment section then identifies threats specific to each of the four empirical studies (the kind-classifier, the user study, the agent comparison, and the rater study), followed by implementation-side concerns and an out-of-scope list. The mitigations the design already carries (three-way inter-rater reliability on the injection sessions, deterministic ranking tie-breaks, multi-knob robustness alongside the single-knob sweep, honest score-floor and validator-check disclosures, reproducible notebook sources for every reported number, and an explicit per-session-not-per-step true-positive policy) are stated where the corresponding design choice or result is presented, rather than restated here.

5.1 Construct validity

There is no external ground truth for “true refactoring quality”. The process-quality score J measures process behaviours the literature identifies as bad practice: leaving the build broken, going long stretches without committing, refactoring manually where an IDE refactor would have worked, and so on. It does not predict bug rates, time-to-merge, or any other downstream-of-refactoring outcome, and Chapter 3 does not claim it does. Calibrating a formula like this against a held-out outcome dataset would require a dataset linking session-level process signals to those outcomes, and we are not aware of one that exists. The weights are grounded in the literature-supported severity orderings of §3.2.2 rather than fitted to data; the chapter addresses robustness and inter-rater agreement empirically, but not predictive validity.

The score is not a maintainability index. Snapshot composites like the Maintainability Index [49] were regression-fitted against expert ratings of completed code. Our score is trajectory-level and process-side; it asks “did the refactoring follow good practice on the way there”, not “is the code good now”. The two are different constructs and we do not conflate them.

The four-kind taxonomy is a methodology choice, not a discovered structure. The divergence-point kinds (*IDE-Replay*, *Rework*, *Hygiene*, *Ordering*) were chosen for the rater-design rationale set out in §3.7: they are operationally distinguishable behaviours the literature flags as bad practice, and they admit a synthesiser that can produce a concrete alternative for each. A different taxonomy is defensible. Collapsing *Hygiene* into separate top-level *Tests-Skipped* and *Commit-Gap* kinds, or merging *Rework* into *Ordering* on the

grounds that both are about wasted steps, would change the precision/recall numbers without changing the chapter’s headline claims.

SuboptimalOrdering is author-claimed, not measured. The five ordering-injection sessions in the playbook carry the `SuboptimalOrdering` pattern label because the playbook author selected an ordering they believed was worse than its alternatives. The score formula cannot independently verify all five: only 1 of the 5 “suboptimal” orderings produces a positive-magnitude reorder alternative under the production formula (§4.2.2, session 022). The other 4 end at zero or negative magnitude, either because the synthesiser could not enumerate any permutations for that window under `splitOnInvalid` (§3.3.2) or because the enumerated permutations tied or scored worse than the user’s order under the production formula. The label reflects what we designed the session to contain rather than objective ground truth, so interpret the precision/recall numbers with that in mind.

The AST-equivalence audit uses an informal definition. A candidate permutation is accepted by the reorder synthesiser only if its terminal AST is equivalent to the user’s. Equivalence is defined as “modulo syntactic differences that do not change behaviour” (§3.3.2) and implemented using an AST hash. The exact set of differences the hash treats as behaviour-preserving (e.g. comment placement, annotation order, whitespace) is implicit in the hash implementation rather than spelled out here. A hash that is too loose would admit reorders that change behaviour and a hash that is too tight would reject reorders that are genuinely equivalent. It catches the obvious failures, but the exact boundary is a design choice we do not formally specify.

5.2 Internal validity

The single-knob sensitivity sweep is combined with a multi-knob design rather than replaced. The headline top-1 stability number in §4.1.1 is computed under single-knob perturbation: one weight changes per row, the rest stay at production. Real mis-specification of weights rarely happens one at a time, so a multi-knob Monte Carlo experiment runs alongside (every weight perturbed simultaneously from $\exp(\sigma \cdot \mathcal{N}(0, 1))$ with $\sigma = \ln 2$). Top-1 stability falls from 96.5% under single-knob to 84.6% under multi-knob on the user-study rankable subset (25 of 30 sessions). The formula is robust to plausibly-sized perturbations of any single weight, and robust to roughly five-in-six plausibly-sized simultaneous perturbations of all weights, but is not robust to arbitrarily-large simultaneous moves. The rank statistics are computed on the rankable subset of each set (15 injection, 25 user-study, 20 agent) because τ -b and top-1 are mechanically 1.0 on sessions with ≤ 1 detected divergence point; the smaller rankable subset on the injection set is a known statistical-power limit on cross-set rank comparison, though the §4.1.2 magnitude-recovery statistics use the full 45 injection sessions where the small-ranking issue does not apply.

Kind-presence agreement does not imply rank-order agreement. The Cohen’s κ reported in §4.2.1 establishes that three independent labellers agree on which divergence kinds fire in each session. It does not establish that they agree on the order the score puts those divergence points in, which is what the dashboard surfaces first to the user. Closing this gap would require a separate rater-rank study (engage raters to rank the divergence points within each session, compute Kendall τ -b against the score’s production ranking). This was out of scope for our project. Our strongest claim from κ alone is that the score’s kind labels match what the rater identifies, but the ranking that orders those labels is not externally validated.

Cleanliness sub-weights are uniform 1.0 by Laplace’s principle of insufficient reason. The six sub-signals that feed the cleanliness sub-score (§3.2.3) are weighted equally because no published calibration data exists for weighting these signals across refactoring trajectories. The sensitivity sweep shows the cleanliness sub-weights have essentially no effect on the top of the divergence-point ranking on the user-study set (Table 4.2: five of six sub-knobs produce zero top-1 disruptions, and `cohesion` produces a single disruption attributable to the lag-denominator coupling). On the agent sessions the cleanliness sub-weights’ Kendall τ drops by 3–10 % under $\times 0.5/\times 1.5$ perturbation – the same coupling running through a smaller session set where one shifted rank dominates. The uniform choice is defensible in practice on the current sessions but it is not a positive calibration result.

The lag term couples the cleanliness sub-score to the process score. The lag term $W_{\text{LAG}} \ell(\tau)$ uses the trajectory’s final cleanliness scalar $C(s_T)$ as the reference against which intermediate-state deficits are measured (§3.2.3). This means a perturbation of any cleanliness sub-weight propagates into the process score through both the endpoint-gain term and the lag denominator, instead of only through the endpoint. The agent-session sensitivity result above is the empirical signature of this coupling. The injection and user-study sessions absorb the same leakage without measurable rank disruption because their per-fixture rankings are longer and more diverse, but the underlying coupling is the same. A cleanliness composite that mis-weights one sub-signal could in principle shift the lag-driven part of the ranking by a small amount we cannot disentangle from the gain-driven part. We treat this as an acceptable design cost: the lag term is meaningful only in the cleanliness-scalar units used elsewhere, and detaching it would introduce a second scalar with no independent calibration anchor.

Two batching primitives are heuristic and not sensitivity-tested. The score formula and the divergence-point detector share two batching constants: `min_commit_gap` = 6 (the green-refactor checkpoint count at which a *Commit-Gap* divergence point fires and the score formula applies the commit-gap penalty), and the 60-second composite-window boundary that bundles refactoring steps into batches for *Tests-Skipped* and *Ordering* detection. Both come from Murphy-Hill et al.’s 2009 study of how developers actually refactor [5], and neither is sensitivity-tested directly. The sensitivity sweep perturbs the weights on these terms but not the thresholds themselves. A different typing speed (slower or faster than the participants in this study) would group steps differently and could produce a different mix of fired kinds.

The production weight vector is hand-picked within a literature-grounded ordering. The exact production weights (§3.2.2, Table 3.2) sit within the severity ordering the literature supports ($W_b > W_{st} > W_{cg}$ etc.), but no calibration step pins them to those specific numbers rather than any other vector inside the same ordering. The robustness experiments establish that the formula sits in a locally stable region around this vector; they do not establish that no other defensible vector exists in the same region.

Which terms carry the ranking varies by session type. The ablation magnitude analysis on the full 45 injection sessions (Table 4.6) identifies `length`, `manualIde`, and `broken` as the strongest magnitude-side contributors (removing each costs sum/sum between 0.18 and 0.28). The cross-set rank-stability analysis on the rankable subsets (Table 4.7) identifies `length` as the dominant rank-carrier on the user-study set ($\tau = 0.527$ on removal), with `manualIde` and `broken` next. The lag term has the smallest user-study rank impact ($\tau = 0.784$ on removal) but a measurable injection-set impact ($\tau = 0.409$), so

its rank-shaping role is narrower on the naturalistic user sessions than on the controlled-injection set. The formula’s effective parameter dimension therefore depends on what kind of session is being scored. This connects to the gain-stripped finding in §4.3.1: the cleanliness-gain term dominates the absolute score but does not reshape the divergence-point ranking on the user-study set, so the formula effectively has two modes (an outcome-side mode that drives the absolute level and a process-side mode that drives the ranking). Whether a weight matters depends on which mode is being read - the absolute score or the ranking.

The reorder dependency analysis is deliberately coarse, and anchors are content-addressed. The dependency analyser that decides which refactoring steps can be safely reordered uses two design choices that affect the *Ordering* recall reported in §4.2.2. First, **Extract*** and **InlineMethod** specs are modelled optimistically: two extracts on the same host method, or two inlines of distinct methods on the same class, are not auto-conflicted, on the grounds that the downstream AST-equivalence check will reject any reorderings that conflict in practice. Second, the four specs that rewrite call sites of a named method (**RenameMethod**, **ChangeMethodSignature**, **ParameterizeVariable**, **ParameterizeAttribute**) are modelled coarsely: each writes the entire declaring class, which chains every later operation on the same class after it by construction, even when the operations would have been genuinely independent. Other rename specs (**RenameClass**, **RenameField**, **RenameLocalVariable**) only consume and produce the specific named entity rather than writing the whole class. On top of this, the entities that anchor each refactoring (a specific region of code, a specific local declaration) are identified by AST subtree hashes, so an identifier rename inside an anchor’s region between two reorderable steps invalidates the second step’s anchor. The optimistic model over-enumerates but the AST-equivalence check filters out the resulting false positives; the coarse model under-enumerates and removes some legitimate independence by construction. The 0.36 *Ordering* recall number reflects these two design choices alongside the **splitOnInvalid** validator check discussed in §4.2.3. Closing the recall gap requires extending the synthesiser engine to reproduce the currently-unsupported step kinds, not refining the dependency analyser; whether the underlying refactorings commute or not is independent of whether the synthesiser can construct the permutations the analyser admits.

Cleanliness sub-signals are clamped to the user’s range on alternative trajectories. Alt-trajectory cleanliness sub-signals are min-max normalised against the main (user) trajectory’s observed range, then clamped to $[0, 1]$. An alternative whose intermediate checkpoints sit outside the user’s range is masked at the boundary. We mention the clamp in §3.2.3 but do not flag it as a threat there; it is named here for completeness. The clamp does not affect the divergence-point magnitudes we report because the validator requires every accepted alternative to reach the same terminal AST as the user’s trajectory, so alt and user share the same terminal-state cleanliness by construction; the trajectory-final J computed from $\Delta C = C(s_T) - C(s_0)$ is identical on both. The clamp only changes how alt intermediate checkpoints are displayed on the dashboard’s cleanliness chart, which has no downstream effect on the scoring numbers.

5.3 External validity

The injection sessions are scoped exercises, not naturalistic sessions. The 45 sessions in the primary test set were recorded by us against a scripted playbook that injects one specific bad behaviour per session. They are scoped exercises rather than naturalistic refactoring activity; precision and recall claims drawn from this dataset are conditional

on that controlled-injection structure, not on the variety of session shapes a free-form developer would produce in practice.

The user study has five participants split into a randomised three-vs-two arm.

Five participants performed six recorded sessions each, 30 in total. Arm assignment was randomised before the study began: two of the five participants were drawn uniformly at random to form the no-feedback baseline arm, and the remaining three form the with-feedback arm. The chapter reports per-participant trajectories and arm-mean comparisons but does not report statistical tests, confidence intervals, or claims like “the median participant improved on metric X ”. The randomised design rules out the alternative reading that the with-feedback arc reflects the playbook getting easier across the arc - the baseline arm worked the same playbook in the same order and trended downward - but $n = 3$ vs $n = 2$ leaves the per-participant variance comparable to the between-arm difference. The strongest claim drawn from the arm comparison is therefore directional, not inferential: under random arm assignment, the with-feedback arm improved across the arc, the baseline arm did not, and the within-arm variance is smaller than the between-arm difference. The remaining confounds after that arm split - demand characteristics and task-style familiarity - are discussed under per-experiment threats in §5.5.

The agent extension is a feasibility experiment, not an agent evaluation.

Eight agent stacks (six with-feedback, two no-feedback baselines) ran the same six-session playbook on the same fixture, 48 sessions in total. The agent corpus exists to observe how the human-trained detectors behave on a structurally different actor, not to evaluate the tool as an agent-monitoring framework. Three structural detector biases against agent traces are documented in §4.3.2 and listed again in the construct-validity item above: agents have no IDE refactor surface, so the *IDE-Replay* kind fires on every structural refactoring without recourse to a fix; the six-checkpoint commit-gap threshold is rarely reached in a six-session arc; and the 2-second edit-burst debouncer collapses each agent turn into one burst. The arm-level ΔJ comparison and the per-stack DP totals are therefore reported as descriptive feasibility evidence rather than as a verdict on the tool’s intended use, which is human-developer refactoring. A multi-stack agent study under a detector set reworked for agent edit patterns is the natural follow-up; that work is named in §5.7 and the conclusion’s future-work section.

All sessions are single-author IntelliJ sessions on a Gradle Java project.

The detector’s behaviour on team pull-request-driven workflows, on other IDEs, or on other languages is not studied. The plugin’s instrumentation is IntelliJ-specific: it uses the IDE’s PSI and VFS hooks to capture refactoring events, which have no direct equivalent in VSCode or Eclipse, so a port would require re-implementing the event-capture layer rather than just retargeting a build.

The detectors are designed for human edit patterns.

The score and divergence-point detectors are built around how human developers interact with an IDE: discrete refactor-menu invocations, edit bursts that the plugin debounces on a 2-second idle, and commit cadences thresholded against the 60-second composite window of [5]. The agent sessions in Chapter 4 are processed by the same pipeline, but the detectors do not map cleanly onto agent traces. Tool-applied file edits collapse onto a single burst because the model’s reasoning pause between turns exceeds the 2-second debounce; the six-checkpoint commit-gap threshold is rarely reached in a six-session arc because each agent turn batches more file changes than a typical human edit; and the *IDE-Replay* kind is structurally moot because the agent has no refactor-menu surface to choose between. The agent comparison

is treated as a feasibility extension on the tooling rather than a target the detectors were tuned for.

Refactoring kinds the synthesiser cannot handle are excluded by construction.

The *IDE-Replay* synthesiser builds alternative trajectories by replaying refactoring specs through Eclipse JDT. A user trajectory built entirely from refactoring operations outside JDT’s supported set (or outside RefactoringMiner’s detection capability, which feeds the synthesiser) would produce no *IDE-Replay* divergence points. The detector cannot flag what it cannot synthesise an alternative for. This is a structural limitation of the detection design, not a finding specific to the sessions we used.

5.4 Statistical conclusion validity

Cohen’s κ in §4.2.1 is reported as a point estimate with raw agreement counts. Confidence intervals are not reported because with only 45 sessions per rater pair, they would be too wide to be useful and would overshadow the κ value itself. Instead, the specific disagreement cases are listed (sessions 024 and 043 for *Rework*; sessions 014, 015, 016, 039 for *Hygiene*), which tells the reader more.

Kendall’s τ -b is brittle on small per-fixture rankings. A single pair swap on a three-point ranking moves τ by 0.333. τ is reported on all three session sets, but the injection sessions’ per-fixture rankings are short enough that the metric is dominated by the small- N artefact; the sweep therefore reports top-1 hit rate alongside τ as a more stable surrogate, and the user-study sessions are treated as the canonical benchmark in §4.1.1.

No multiple-comparison correction is applied across the weight $\times$ factor sweep ($12 \times 9 = 108$ perturbation cases per fixture). The headline result is a descriptive statistic - what fraction of cases preserve the top-1 recommendation - not a hypothesis test, so a correction would be appropriate only if individual cases were being interpreted as significance claims, which they are not. The multi-knob Monte Carlo uses a fixed seed and the 200-sample size was chosen heuristically rather than from a power calculation; distribution shape (mean τ , top- N hit rate) is reported rather than a point estimate with confidence interval.

5.5 Per-experiment threats

The kind classifier (§4.2.2). The 1.00 precision figure across all four kinds is conditional on the sessions used. `expected_kinds` is a hand-label computed independently of the playbook’s injection pattern, and the inter-rater κ study (§4.2.1) establishes that three labellers agree on it for these sessions. The threat is not that the labels are author-biased; it is that the dataset consists of controlled-injection sessions whose divergence behaviours are bounded by what the playbook injected and what the four detector kinds can describe. A free-form session set could contain bad behaviours outside those four kinds, and even within the four kinds the labelling shows rater variance on this dataset ($\kappa = 0.72$ on *Hygiene* between two of the three pairs, §4.2.1). Precision on a free-form session set is therefore unstudied. Separately, IDE-replay precision was 0.80 before the plugin-instrumentation fix described in §4.2.3: the false positives in the pre-fix recordings were upstream plugin-capture issues, not detector-logic errors, but the current 1.00 figure would not hold on the pre-fix recordings.

The user study (§4.3.1). The external-validity limitations above ($n = 3$ vs $n = 2$, randomised arm assignment) apply here too. The metrics in the gain-stripped score (commit cadence, skipped tests, manual-when-IDE rate) don’t depend much on knowing the codebase, so the with-feedback arm’s improvement can’t be explained by just getting more

familiar with the fixture - and the baseline arm worked the same fixture on the same playbook and trended downward, which removes “the playbook got easier as the participant progressed” as a competing explanation. Two confounds remain after the arm split. First is the Hawthorne effect: after seeing their dashboard in S1, a with-feedback participant figures out what the system rewards (frequent commits, test runs, IDE actions) and adjusts their behaviour to satisfy it. This looks like improvement, but it is behaviour change to fit the metric, not skill improvement. We can’t tell the two apart from the data. Second is learning the task format: all six sessions follow the same structure (a what-not-how prompt, a small change to make, look at the dashboard), so with-feedback participants get better at the format itself, not at refactoring. The baseline arm’s downward trend across the same six sessions makes this second explanation harder to sustain in isolation - if format-familiarity alone drove the with-feedback improvement, the baseline participants would also have improved as the format got easier - but it does not formally rule it out. We do not claim either confound is ruled out by the arm split.

The agent extension (§4.3.2). The agent extension is a feasibility experiment for the tool on a structurally different actor, not an agent-monitoring evaluation. Three of the four detector kinds structurally under-fire on agent traces (§4.3.2): IDE-replay fires on every structural refactoring an agent performs because the agent has no IDE refactor surface to switch to, the commit-gap threshold is rarely reached in a six-session arc, and the edit-burst debouncer collapses each agent turn into one burst. The null arm-level ΔJ result in §4.3.2 reflects detector scope as much as agent behaviour, and would need to be revisited under a detector set reworked for agent edit patterns before any quantitative agent claim could be made. The per-stack engagement split between Claude-family and non-Claude stacks is reported as a qualitative observation from the transcripts, not as a finding about the tool’s causal effect. With $n = 6$ vs $n = 2$ across the arm comparison and one stack per axis position in the cross-stack comparison, no inferential weight is given to the per-stack figures.

The rater study (§4.2.1). Three labellers, one of whom is us. The chapter reports pairwise κ so us-vs-rater values are visible alongside the rater-vs-rater pair, which makes our self-bias detectable rather than hidden. The two participant raters are the same two participants who later performed the user-study sessions, so familiarity with the broader project is not zero; the labels were assigned before any participant performed a user-study session, however, so the labelling cannot have been biased by what the participants later did. Two of the four kinds (*Rework* and *Hygiene*) show $\kappa < 1.0$ with specific disagreement structure named in §4.2.1; reconciliation was deliberately not applied to inflate κ , since the disagreements are themselves a finding about where the kind boundaries are fuzziest.

5.6 Implementation threats

The wrap-and-patch layer of §3.3.3 closes residual gaps between IntelliJ and Eclipse JDT for accepted reorder windows but is not exhaustive: windows where the gap cannot be patched are marked `ast_diverged` and excluded from reordering.

The reorder enumerator carries a hard size budget: any window with more than 7 refactoring steps is skipped entirely, and within an admitted window enumeration stops after $7! = 5040$ permutations (`TopologicalEnumerator.kt`). The budget exists because the cost of enumerating all topological orderings grows factorially with window size, and an 8-step window would already require $\geq 40,320$ enumerations before the AST-equivalence audit ran on each. In practice the recorded sessions rarely produced reorder windows large enough to exceed this budget, but a session with longer contiguous refactor sequences

would have its widest windows silently dropped from the *Ordering* enumeration. A future implementation could replace the exhaustive enumeration with a beam search or heuristic pruning to lift the size cap.

5.7 Out of scope

The following are explicitly outside the scope of our work. We do not claim to address them.

- **Agent-side detector rework.** The agent extension of §4.3.2 runs the existing human-trained detectors on eight agent stacks. A quantitative agent evaluation would first require reworking the detectors against an agent-edit model (tool-call grouping in place of edit-burst debouncing, per-turn commit accounting in place of the six-checkpoint window, and an agent-tool-availability check in place of the IDE refactor surface). That work is named in Chapter 6.
- **Longitudinal validation against external code-quality outcomes.** No outcome dataset linking refactoring-trajectory signals to downstream maintenance cost was available, and constructing one was treated as outside our scope.
- **Cross-IDE and cross-language validation.** The plugin is IntelliJ-specific and the refactoring engine is JDT-specific (Java). Ports to other IDEs or languages are named as future work and not attempted.
- **Production-weight refitting against an outcome dataset.** See the outcome-dataset point above.
- **Real-time, live-feedback dashboard mode.** The dashboard is a session-replay view of completed analysis reports. Live feedback during refactoring is flagged in §4.3.1 as a natural next step but is not implemented.
- **Rater-rank validation.** As named in §5.2, the κ study validates kind-presence agreement, not rank-order agreement. A Kendall τ -b study on per-session rankings is the natural follow-up and is not done here.

Chapter 6

Discussion and Conclusions

This chapter consolidates the work of the previous chapters: what was built, what Chapter 4 establishes and what it does not, and which directions are most worth pursuing next.

6.1 Summary of contributions

We produce two main contributions. The first is the process-quality score J , a formula that measures refactoring quality by combining the endpoint cleanliness gain with five penalty terms from the literature – broken builds, skipped tests, manual edits when IDE refactoring would work, long gaps between commits, and an intermediate-cleanliness lag penalty that costs trajectories which linger below their final cleanliness target – and a step-savings bonus that rewards alternatives reaching the same end state in fewer steps. The second is the divergence-point detector, which finds points in a refactoring session where a better alternative exists, classifies them into four kinds (*IDE-Replay*, *Rework*, *Hygiene*, *Ordering*), and produces a concrete alternative trajectory for each. The alternatives are generated by four per-kind synthesisers internal to the detector: a reorder enumerator that finds valid step reorderings and checks them for AST equivalence, an IDE replayer that converts manual edits into IDE refactor calls, a rework stripper that removes self-cancelling step pairs, and a hygiene synthesiser that models what would happen if tests were run on commits made at different points.

Supporting these are: a 45-session hand-recorded dataset on a purpose-built Java fixture with multi-label ground-truth annotations; a 30-session randomised-arm user study (five participants split into a with-feedback arm of three and a no-feedback baseline arm of two) on a separate fixture; an 8-stack, 48-session agent extension run on the same fixture as a feasibility experiment for the tool’s human-edit-pattern detectors; the IntelliJ plugin that captures the event traces; the JCEF dashboard that surfaces the divergence points; and a reproducible notebook from which every number in Chapter 4 can be regenerated.

6.2 Headline findings

Four findings carry our main claims.

First, the kind classifier is precise on the controlled-injection sessions. Per-kind precision is 1.00 across all four kinds against the inter-rater-agreed `expected_kinds` labels, with Cohen’s $\kappa = 1.00$ for *Ordering* and *IDE-Replay* on all three rater pairs and $\kappa = 0.72$ –1.00 for *Rework* and *Hygiene* (§4.2.1, §4.2.2). Every injection caught with a positive-magnitude alternative trajectory ranked at top-1 within its session.

Second, the score formula sits in a locally stable region around its production weights. Top-1 stability under single-knob perturbation is 96.5% on the rankable subset of the user-

study set (25 of 30 sessions; the remaining sessions have ≤ 1 detected divergence point and are excluded from the rank statistic because both top-1 and Kendall τ -b are mechanically 1.0 on them). The multi-knob Monte Carlo design (every weight perturbed simultaneously from $\exp(\sigma \cdot \mathcal{N}(0, 1))$ with $\sigma = \ln 2$) lowers this to 84.6% (§4.1.1). The formula is robust to plausibly-sized single-weight changes and to a majority of plausibly-sized simultaneous changes, and degrades smoothly under larger ones.

Third, the user-study arm comparison shows a clear direction of effect. With-feedback participants gain +21.0 on the gain-stripped process score across the six-session arc (mean S1 26.7 $\rightarrow$ S6 47.7) and drive total divergence points from 12 to 2; no-feedback baseline participants lose 5.0 (mean S1 22.5 $\rightarrow$ S6 17.5) and rise from 10 to 15 DPs. With $n = 3$ vs $n = 2$ the result is randomised but underpowered, so no p -value is claimed; the within-arm variance is comparable to the between-arm difference and the finding is reported as a direction-of-effect signal. The two arms start at comparable totals at S1 and diverge from S3 onwards, which rules out the alternative reading that the with-feedback arc reflects the playbook getting easier across the arc (§4.3.1).

Fourth, the agent extension (eight agent stacks across model $\times$ harness combinations, six with-feedback and two no-feedback baselines, 48 sessions total) is a feasibility experiment for the tool on a structurally different actor rather than a co-equal evaluation. The tool’s human-edit-pattern detectors structurally under-fire on agent traces: across 48 agent sessions the corpus contains 42 divergence points, 41 of which are IDE-replay, because agents have no IDE refactor surface to switch to and three of the four kinds barely fire on agent traces at all (§4.3.2). On this corpus the between-session feedback shows no measurable causal effect on agent process scores; both arms’ slopes are positive but the baseline arm’s slope (+1.7) is comparable to the with-feedback arm’s (+0.8), and what difference appears runs in the opposite direction to the human result. Per-stack transcripts add a more interesting qualitative finding: the three Claude-family stacks visibly engaged with the feedback in writing (stack 1’s commit cadence revisions, stack 2’s atomic-edit synthesis, stack 3’s S5 IDE-replay adjustment), while the three non-Claude stacks (Gemini, GPT-5.5 Cursor CLI, GPT-5.5 OpenCode) showed no visible engagement; the model-family split is descriptively interesting but does not rescue a causal claim on the gain-stripped score.

6.3 Discussion

The chapter substantiates four claims: a divergence-point classifier whose precision and rank-1 prominence are conditional on the recorded sessions but reproducible; a score formula whose ranking is locally robust to weight choices, with each term carrying measurable influence; a randomised but underpowered direction-of-effect comparison in which the with-feedback user-study arm gained +21.0 on gain-stripped process score across the six-session arc while the baseline arm lost 5.0; and an agent extension experiment whose null result on the same arm comparison is consistent with the tool’s human-edit-pattern detectors structurally under-firing on agent traces.

A few stronger claims are left for follow-up work. The user-study arm comparison is randomised but small ($n = 3$ vs $n = 2$), so the direction of effect is reportable but no p -value or confidence interval is claimed; the threats chapter (§5.1) discusses the demand-characteristics and task-familiarity confounds that remain. The agent extension is treated as a feasibility experiment rather than a co-equal evaluation: three of the four detector kinds rarely fire on agent traces by construction, so the null arm-level finding for agents reflects detector scope as much as agent behaviour, and a quantitative agent claim would require reworking the detectors against an agent-edit model first. The plugin and synthesisers are written against IntelliJ and JDT, so generalisation to other IDEs or languages is a port rather than a configuration change. The weights are grounded in literature-supported

severity orderings rather than fit against an external outcome dataset, since no dataset linking session-level process signals to downstream maintenance outcomes exists at the granularity the score operates on. Chapter 5 discusses each of these in detail.

6.4 Future work

The suggestions below are ordered by what would most strengthen the main findings, with the scale of the human study the largest single gap:

1. **Larger randomised human study.** The strongest causal-direction claim in the chapter (the with-feedback vs no-feedback arm comparison in §4.3.1) is randomised but underpowered at $n = 3$ vs $n = 2$, with per-participant variance comparable to the between-arm difference. A study with more participants in each arm would let the direction-of-effect signal become a hypothesis test rather than a descriptive claim, and would allow proper estimation of between-participant variance.
2. **Adapting the tooling to agent traces.** The detectors in this thesis are designed against human edit patterns. The agent comparison in §4.3.2 is a feasibility extension on the same tooling rather than a co-equal evaluation, because three of the detector kinds (*IDE-Replay*, *Hygiene* via the commit-gap window, and edit-burst structure) bias against how agents emit file changes. Reworking those detectors against an agent-edit model – tool-call grouping in place of edit-burst debouncing, per-turn commit accounting in place of the six-checkpoint window, and a tool-availability check in place of the IDE refactor surface – would let the same scoring framework produce results for agents that the current detectors cannot.
3. **Rater-rank validation.** The Cohen’s κ study validates kind-presence agreement only. A Kendall τ -b study on per-session divergence-point rankings would validate that the score’s ordering of divergence points within a session matches a human expert’s ranking, which the κ figures alone do not establish.
4. **IDE-refactor-as-agent-tool deployment.** The IDE-replay structural limitation in §4.3.2 is the cleanest example in the chapter of feedback being correctly interpreted but inactionable: the agent acknowledges the IDE refactoring would score better but has no way to invoke it. Exposing IntelliJ’s refactor actions as agent tool calls would close that gap and would let the *IDE-Replay* count actually respond to feedback across sessions.
5. **Extend the reorder synthesiser to lift Ordering recall.** §4.2.3 documents the 0.36 Ordering recall as a real limit: with the lag term W_{LAG} in the production formula, commuting permutations can have different magnitudes, so the windows currently rejected by `splitOnInvalid` would contribute non-zero magnitudes if the synthesiser could replay them. Extending the synthesiser engine to reproduce the currently-unsupported step kinds would directly lift recall and surface additional positive-magnitude Ordering divergence points.

6.5 Closing

The introduction opened by noting that the literature evaluates refactoring as a mapping from a start state to an end state, with the trajectory between them largely outside the picture. Our work shifts the question being asked: from whether the code is better after the refactoring to whether the process the developer followed is one a careful developer would have followed. The point underneath the numbers in Chapter 4 is that a refactoring

trajectory has concrete alternative paths at identifiable points, and that the gap between the observed path and those alternatives is not something a snapshot-only evaluation can see.

Chapter 7

Declarations

7.1 Use of Generative AI

I acknowledge the use of Claude Code (Anthropic, <https://claude.com/claude-code>) and the underlying Claude Opus / Sonnet models while implementing the tool described in this thesis. AI assistance was particularly useful for these development tasks, under my direct supervision:

- *Repetitive adapter and wrapper code.* The analysis pipeline bridges IntelliJ’s recorded refactoring events to Eclipse JDT’s refactoring API. Many refactoring specs, such as `RenameMethod`, `RenameField`, `RenameClass`, `RenameLocalVariable`, `ExtractMethod`, `InlineMethod`, `ChangeMethodSignature`, and `ParameterizeVariable`, required similar wrappers with small differences in fields, preconditions, and follow-up validation. After writing the first versions and fixing the shared interface, validation logic, and AST-equivalence check, I used AI assistance to draft further wrappers following the same pattern.
- *Serialisation and test scaffolding.* AI assistance was used for Kotlin data classes for the analysis-report JSON schema and for fixture-builder code in the test suite.
- *Understanding unfamiliar APIs.* I used AI to help explain parts of the IntelliJ Platform plugin SDK, the Eclipse Equinox bootstrap, and the JCEF dashboard environment, especially where the official documentation was limited.
- *Fixture and playbook ideas.* AI was used to help generate ideas for the Java fixtures and scripted playbooks used in the manual-injection, user-study, and agent-comparison sessions. These were purely test data. I edited the scenarios manually, and verified that they exercised the intended behaviours.

All AI-assisted code was reviewed manually. The analysis, refactoring, and dashboard modules are also covered by tests, which provide a separate check on the behaviour of the implementation.

For the report itself, the content, argument, and findings are my own. Claude (<https://claude.ai>) was used for proofreading, structural feedback on draft text, and help with L^AT_EX formatting for tables and figures. No AI-generated content has been presented as my own independent work.

7.2 Ethical Considerations

The project includes a small user study with five human participants, referred to as P1–P5, and a separate agent comparison with no human subjects. All participants were senior

MEng Computing students at Imperial College London. They consented to having their sessions recorded and analysed, on the basis that no personally identifying information would appear in the report or released materials. The participant traces and captured edits are pseudonymised, and the underlying recordings are kept under access controls in the project repository.

The project fits Imperial’s description of low-risk research, as it involved only non-sensitive session data from consenting participants and posed no risk beyond ordinary development work [59]. No formal ethics approval was required. It did not use medical data, sensitive personal data, or third-party-owned datasets.

7.3 Sustainability

The project’s compute footprint was kept small by two design choices described in the methodology chapter:

1. *Phase A / Phase B pipeline split.* The expensive analysis work, including shadow-repository reconstruction, RefactoringMiner mining, and build/test execution at each checkpoint, is run once per recorded session and cached. Later scoring, sensitivity, ablation, and Monte Carlo experiments reuse this cached output and rerun only the cheaper scoring phase. This avoids repeating the full analysis for every weight configuration.
2. *Deterministic ranking and reuse across runs.* The experiment drivers use fixed random seeds and deterministic tie-breaking in the divergence-point ranking. As a result, rerunning an experiment for a sanity check costs no more than the original run and produces the same CSVs when the inputs have not changed.

All experiments were run on a single local workstation. No GPU or cloud-compute resources were consumed.

7.4 Availability of Data and Materials

The project materials are available as follows:

1. **Source code:** <https://github.com/EthanHosier/fyp>. The repository contains the IntelliJ plugin, analysis pipeline, JCEF dashboard, and experiment drivers.
2. **Recorded sessions, fixtures, and supporting material:** the recorded sessions, playbook prompts, fixtures, and session-recording protocol are stored under `tool/fixtures/`. This includes the 45 hand-recorded injection sessions, 30 user-study sessions across 5 participants, and 48 agent-comparison sessions across 8 agent configurations. The directory layout is documented in `tool/fixtures/README.md`.
3. **Reproducibility of Chapter 4:** the numerical claims, tables, and plots in Chapter 4 can be reproduced with `tool/fixtures/notebooks/experiments.ipynb`. The notebook reads the per-session cached analysis files and produces the reported aggregates. Setup instructions are provided in `tool/fixtures/notebooks/README.md`.
4. **Participant data:** participant data is pseudonymised. No personally identifying information is included in the repository or in this report.

Appendix A

Appendix

This appendix is intentionally left available for supplementary material that is useful to include in the submitted report but too detailed for the main chapters.

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